
Continuous Audits of Representation and Alignment Uncertainty in Cryo-EM

Fourier Cryo Splats Project¹

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Abstract

Uncertainty in a cryogenic electron microscopy (cryo-EM) reconstruction depends on both its representation and its treatment of estimated alignments. We develop a continuous Fourier-slice audit for fixed linear density features, separating a convenient reconstruction model from the class over which uncertainty is assessed. Analytic adjoint integrals, matrix-free optimization, and explicit integration remainders permit an audit outside a Gaussian fitting dictionary. We distinguish deterministic pose errors from a conditional model of independent centered errors with a bounded shared component, and show why independent inference images alone do not establish the latter model. A matched continuous Gaussian-prior interval can share the fixed-pose uniform guarantee, so fixed-pose coverage is not an advantage over that baseline. A frozen study re-estimates local poses and weights on 600 noisy datasets across three acquisition geometries. All 216 procedure/control cells cover their broad targets in 200/200 trials, including the simpler fixed-pose and Gaussian baselines. The three pose-audit procedures always select the no-data fallback; their raw median widths are 642–1,242 times its width. A post hoc replay finds alignment bias hidden by these wide intervals: noise-only coverage drops to .72 for one pilot-aligned feature, versus .955 with true poses. Exact continuous bias envelopes at realized poses distinguish bound looseness from class sensitivity. Experimental interval centers were observed before fresh noise calibration; density, pose, and covariance assumptions remain unvalidated. This manuscript does not establish experimentally calibrated density coverage or an advantage at near-atomic resolution.

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1. Introduction

The relevant question for an uncertainty method is which claim its interval supports. A shell correlation, a posterior over conformations, and a local density confidence interval have different estimands and sampling assumptions. Modern neural and Gaussian reconstructions can represent detailed densities, but representational flexibility alone does not determine which features are supported by the observations. We study a specified local average or contrast of a homogeneous density, including the effect of uncertainty outside the representation used to fit a pilot.

Two distinctions motivate the method. First, conditioning on a finite reconstruction dictionary removes possible density errors by assumption. An independent-half reconstruction can inherit the same omission. Second, an interval derived for a fixed Fourier-slice design need not remain valid after estimating that design from noisy particles. Separating alignment and inference observations removes one noise dependence, but does not make alignment errors centered conditional on the resulting design.

Our contribution is an implemented continuous Fourier audit and a validation program that exposes these distinctions quantitatively. It specializes established bias-aware inference: the statistical coverage principle, norm duality, and Gaussian ridge representation are not new. The operator-specific work evaluates continuous residual and pose-derivative fields, controls quadrature in real arithmetic, and records when the resulting intervals are informative. The full study retains matched favorable baselines, numerical failures, and no-data fallbacks. A conditional guarantee is kept separate from evidence that its physical assumptions hold.

2. Related Work and the Uncertainty Target

Bayesian reconstruction is foundational in cryo-EM (Scheres, 2012). Posterior ensembles from Gaussian pseudo-atoms (Joubert & Habeck, 2015), variational Fourier-slice density uncertainty (Ullrich et al., 2020), and likelihood-Hessian soft modes (Rangan et al., 2024) directly precede our problem. The latter two already distinguish variance from model error and expose density–pose coupling. We

compare against a continuous Gaussian prior with the same operator and directional scale rather than treating a weaker discretized prior as decisive evidence.

CryoDRGN (Zhong et al., 2021) and ab initio neural reconstruction (Levy et al., 2025) address representation learning and structural recovery. Physical heterogeneity is not automatically uncertainty about one homogeneous density. Gaussian representations are also established (Chen & Ludtke, 2021; Chen & Ling, 2026; Qu et al., 2025; Shekarforoush et al., 2025; He et al., 2026); their existence is not our novelty claim. Gaussian Fourier kernels provide our initial pilot, while the final audit does not require a Gaussian density dictionary.

Gold-standard halves and noise substitution assess overfitting (Scheres & Chen, 2012; Chen et al., 2013); confidence maps test signal significance (Beckers et al., 2019; 2020); BioEM and independent-particle validation assess image fit (Cossio & Hummer, 2013; Ortiz et al., 2020). Enhancement confidence can be calibrated against a proxy map (Bharadwaj et al., 2026), and conformal methods can cover an exchangeable population of signal–observation pairs (Narnhofer et al., 2024). None should be scored as though it promised our fixed-signal density guarantee. The survey in Appendix F separates these targets and identifies unresolved research questions.

Resolving kernels (Backus & Gilbert, 1968), optimal recovery (Donoho, 1994), and honest bias-aware intervals (Armstrong & Kolesár, 2018) supply the statistical foundations. GP posterior variances and worst-case kernel errors have established connections (Kanagawa et al., 2025); frequentist GP bounds also require explicit norm and noise assumptions (Fiedler et al., 2021). Moment methods can recover collective information without estimating every pose (Sharon et al., 2020; Hoskins et al., 2026). Therefore, failure of a local pose audit is not an impossibility result for cryo-EM reconstruction.

3. Model, Feature, and Conditional Design

For Fourier convention $F(k) = \int \rho(x) e^{-2\pi i k^\top x} dx$, write a particle as

$$Y_i(q) = C_i(q) e^{-2\pi i q^\top t_i} F(E_i q) + \epsilon_i(q). \quad (1)$$

E_i embeds the detector plane, C_i is the contrast transfer function (CTF), and frequency and shift units are reciprocal. We realify one representative of each conjugate pair. After specified whitening, the conditional nominal model is

$$y = \mathcal{A}_0 \rho + d + \epsilon, \quad \epsilon \sim N(0, I), \quad (2)$$

where d represents pose or other imaging error. The feature is $t = \langle \ell, \rho \rangle$, with a stated coordinate frame and physical

smoothing width. We use a Gaussian local average or signed difference of two averages.

Let $H = L^2([-1/2, 1/2]^3)$ and let an independent pilot ρ_0 define $\delta = \rho - \rho_0$, $\|\delta\|_H \leq B$. This supported continuous class is different from a finite voxel or Gaussian class. The norm radius is an assumption, not a consequence of a small data residual. The estimator is

$$\hat{t} = \langle \ell, \rho_0 \rangle + w^\top (y - \mathcal{A}_0 \rho_0). \quad (3)$$

For the basic theorem, the pilot, design, target, noise model, and weights are fixed independently of inference noise. Using consensus poses fitted to the same particles does not satisfy that premise. If separate alignment images determine the design, the theorem can instead be conditioned on those images, provided the inference noise and its processing have the specified conditional law.

4. Continuous Representation Audit

Put $h = \ell - \mathcal{A}_0^* w$, $s = \|w\|$, and suppose $|w^\top d| \leq b_{\text{pose}}$. The total bias bound is

$$b = B \|h\|_H + b_{\text{pose}}. \quad (4)$$

Let $q(s, b, \alpha)$ solve

$$\Phi((q - b)/s) - \Phi((-q - b)/s) = 1 - \alpha \quad (5)$$

for $s > 0$, and set $q(0, b, \alpha) = b$.

Theorem 1 (Fixed-design coverage). *Under Equation (2), with density and nuisance error fixed independently of inference noise, $\hat{t} \pm q(s, b, \alpha)$ covers t with probability at least $1 - \alpha$, uniformly over the declared density and nuisance class.*

Proof. The error is $-\langle h, \delta \rangle + w^\top d + w^\top \epsilon$. Its fixed absolute mean is at most b , and its Gaussian standard deviation is s . Symmetric Gaussian coverage decreases with absolute mean; Equation (5) evaluates its minimum at b . $\square$

The theorem is a classical bias-aware construction. A bias chosen adaptively as $b \text{sign}(w^\top \epsilon)$ need not obey its folded-normal conclusion. The sum width $b + z_{1-\alpha/2}s$ handles a bounded adaptive bias when the noise projection itself remains fixed Gaussian, but not arbitrary data-dependent weights.

4.1. Auditing outside the fitting dictionary

If a reconstruction space has orthogonal projector P_S , a restricted calculation uses only $\|P_S h\|$. Orthogonality gives

$$\|h\|^2 = \|P_S h\|^2 + \|(I - P_S)h\|^2. \quad (6)$$

The second term is discarded uncertainty, even if the restricted residual is zero. Our audit evaluates the full continuous residual without estimating an omitted density direction from the inference images. It remains conditional on the declared support and norm class.

On the unit cube, $K(v) = \prod_a \text{sinc}(v_a)$. With $c_j = (w_{Rj} + iw_{Ij})C_j/\sigma$ and $f_w(x) = \text{Re} \sum_j c_j e^{2\pi i k_j^\top x}$,

$$\|f_w\|^2 = \frac{1}{2} \text{Re} \sum_{j,l} \{c_j \bar{c}_l K(k_j - k_l) + c_j c_l K(k_j + k_l)\}. \quad (7)$$

Analytic Gaussian target integrals supply the remaining terms of $\|\ell - \mathcal{A}_0^* w\|^2$. A matrix-free implementation quadratures only this known adjoint field, not the unknown density. Explicit integration remainders accompany primal residuals and dual feasibility. FINUFFT and rounding errors are checked numerically, not enclosed by interval arithmetic (Appendix B).

4.2. Design and the continuous Gaussian baseline

We minimize the convex sum objective

$$z_{1-\alpha/2} \|w\| + B \|\ell - \mathcal{A}_0^* w\| \quad (8)$$

before evaluating the sharper folded-normal width. At an interior solution, stationarity gives $(G + \lambda I)w = a$, with $\lambda = zb/(B^2 s)$ for $b = B \|h\|$. Thus these weights lie on the same ridge path as the Gaussian comparator; the comparison changes regularization and critical value, not the estimator family. A feasible dual supplies a stopping gap for this objective; it does not establish optimality of the reported folded-normal width. A no-data alternative uses center $\langle \ell, \rho_0 \rangle$ and half-width $B \|\ell\|$. If selected using only design information, both center and width are replaced.

Write $G = \mathcal{A}_0 \mathcal{A}_0^*$ and $a = \mathcal{A}_0 \ell$. A continuous isonormal Gaussian prior of directional scale τ gives

$$w_\tau = (G + \tau^{-2} I)^{-1} a, \quad (9)$$

$$v_\tau = \|w_\tau\|^2 + \tau^2 \|\ell - \mathcal{A}_0^* w_\tau\|^2. \quad (10)$$

White covariance is not trace class on infinite-dimensional L^2 : this defines joint Gaussian observations and functionals, not an L^2 -valued random density. It is the continuous ridge/GP comparison naturally associated with the audit.

Proposition 1 (Fixed-bias Gaussian envelope). *For $z \geq \sqrt{3}$ and $t \geq 0$, $\Pr(|N(t, 1)| \leq z\sqrt{1+t^2}) \geq 2\Phi(z) - 1$.*

The proof is in Appendix A. At $\tau = B$, $z_{1-\alpha/2}\sqrt{v_\tau}$ therefore shares the uniform fixed-pose class guarantee for $\alpha \leq 2\Phi(-\sqrt{3})$. The folded-normal critical value can be shorter; uniform coverage alone is not a distinction from this baseline. We also retain $\tau = B/2$, for which that

implication need not hold. A pose-linearized comparator replaces identity noise covariance by $I + JJ^\top$ using the actual continuous pilot Jacobian. It omits nonlinear and unknown-density/pose interactions.

5. Pose Errors and What Splitting Changes

For particle i , let $D_i : \mathbb{R}^5 \rightarrow H$ be the first derivative of its weighted adjoint field with respect to scaled rotation/translation error. Its continuous 5×5 Gram is obtained by differentiating the sinc kernel. The scalar derivative satisfies

$$\|D_i^* \rho\| \leq c_i := \|D_i^* \rho_0\| + B \|D_i\|_{\text{op}}. \quad (11)$$

Let R_2 be a uniform scalar second-order remainder over the specified error radii, computed by integrated Fourier moment bounds (Appendix C). For arbitrary bounded errors $\|u_i\| \leq r_i$, a deterministic bias bound is $B \|h\| + \sum_i r_i c_i + R_2$. This term need not average away as particle count increases.

An alternative model writes $u_i = \mu_i + \xi_i$, with $\|\mu_i\| \leq \eta_i$ and independent centered ξ_i satisfying $\|\xi_i\| \leq r_i$, conditional on the frozen design, weights, density, and pilot. The ξ_i are independent of measurement noise. This is a stronger assumption than a bounded pose radius.

Theorem 2 (Conditional mixed-pose bound). *Under this model, the interval centered at Equation (3) with half-width*

$$B \|h\| + \sum_i \eta_i c_i + R_2 + \sqrt{2 \log(2/\alpha) \left(\|w\|^2 + \sum_i r_i^2 c_i^2 \right)} \quad (12)$$

has coverage at least $1 - \alpha$. The remainder is evaluated over radii $r_i + \eta_i$.

Proof. The first-order random scalar for particle i is centered and lies in $[-r_i c_i, r_i c_i]$. Hoeffding's lemma and independence add their sub-Gaussian variance proxies to the Gaussian measurement variance $\|w\|^2$. The two-sided Chernoff bound gives the square-root term. Density bias, shared error, and nonlinear remainder are bounded in absolute value and added outside that tail event. $\square$

We use a sub-Gaussian critical value; substituting a Gaussian quantile would require an additional approximation or distributional assumption.

Estimated poses do not establish conditional centering.

For fixed true poses, a local error $u = \theta - \hat{\theta}$ is measurable with respect to a design containing $\hat{\theta}$. Hence $E[u \mid \hat{\theta}] = u$, not generally zero. Unconditional symmetry does not

supply the premise of Theorem 2. A random latent-pose model or an unconditional analysis of coupled errors and weights needs its own argument. Our estimated-pose use of Equation (12) is accordingly an empirical plug-in test, not an automatic application of the theorem.

The issue remains with independent inference images. In the phase model $A = s + \sigma_A Z$, $B = s + \sigma_B W$, with independent circular Gaussian noise and $s > 0$, let $\hat{\theta} = \arg A$. Same-image alignment has mean $E|A| > s$, whereas independent-image alignment has mean $E \operatorname{Re}(e^{-i\hat{\theta}} B) = sE \cos \hat{\theta} < s$. Their shrinking intervals can converge to zero coverage. This elementary control and a finite-sample invariant comparator are detailed in Appendix D; alignment attenuation and cross-validation are established concerns.

6. Validation

We separate three evidence levels: conditional mathematical/numerical checks with supplied inputs; simulations that repeat estimation under a known generator; and experimental-image applications with unverified inputs. The unit of replication is a newly generated dataset or a genuinely independent exposure, not a target, a method, or multiple intervals on the same data.

6.1. Acquisition geometries and estimands

We use EMPIAR-10028 (80S ribosome), 10049 (RAG complex), and 10076 (ribosomal assembly intermediates). Their working fields are 482.4, 236.16, and 419.2 Å. Density generators use the frozen pilot-registered EMD-2660, EMD-6487, and EMD-8434 maps. The last is one Class A state, not truth for the heterogeneous 10076 stack. Source-group splits, exact selected indices, download records, and input hashes are retained.

The local-refinement study uses 128 particles, frequency radius five, and two broad Gaussian targets of width .07 field. The separate locked-feature study uses radius twelve and twelve features of width 20 Å. Band endpoints are 47.2–96.5 Å and 19.7–40.2 Å, respectively. These experiments address coarse features, not near-atomic accuracy. Supplied scalar noise is used in simulations; experimental recalibration is separate.

6.2. Representation and matched-prior checks

All twelve original finite-dictionary estimators replay their archived metrics before changing the generator frame. For the central broad 10028 feature, restricted analytic coverage is 4.18×10^{-5} before registration and 3.73×10^{-14} afterwards. Other features improve or worsen; the full table is retained. Continuous/ambient bias checks address the omitted-space mechanism, but conservative coverage

Table 1. Matched continuous Gaussian comparison: range over four locked features per stack. R is half-width divided by the original fixed-pose audit width. $C_{\min}$ is minimum analytic coverage across both generator frames and all seven prescribed pose settings. All 48 solves converge; this is conditional supplied-noise coverage, not refitted-pose validation.

Stack	Pose model	τ	R range	$C_{\min}$
10028	Fixed	2	1.341–1.353	1.000
10028	Linearized	2	1.342–1.354	1.000
10028	Fixed	1	0.874–0.876	1.000
10028	Linearized	1	0.876–0.877	1.000
10049	Fixed	2	0.994–0.994	0.996
10049	Linearized	2	0.994–0.994	0.996
10049	Fixed	1	0.983–0.984	0.996
10049	Linearized	1	0.983–0.984	0.996
10076	Fixed	2	0.999–0.999	0.996
10076	Linearized	2	0.999–0.999	0.996
10076	Fixed	1	0.990–0.990	0.996
10076	Linearized	1	0.990–0.990	0.996

of these particular generators does not calibrate the density class.

The matched continuous Gaussian comparison includes all twelve locked features, $\tau \in \{B, B/2\}$, and fixed/linearized-pose models: 48 fits. Both original and registered generators and all declared perturbations are included. Non-convergence is reported rather than interpreted as an exact posterior. A numerical-only preflight motivated stronger preconditioning for a separate complete rerun; the original incomplete implementation and failed fits remain available. All 48 rerun fits converge. At $\tau = B$, half-width ratios to the fixed-pose audit are 1.341–1.354 on 10028, .994 on 10049, and .999 on 10076. Minimum conditional coverage across all 672 generator/scenario cases is .995970 against the nominal .995833 level, including the half-scale prior. These favorable fixed-generator results do not supply a uniform guarantee for the half-scale prior or for estimated poses.

6.3. Re-estimated local poses and weights

An independent calibration batch contains 128 datasets per geometry. Test batches contain 200 new datasets per geometry. On every replicate, ten fixed Gauss–Newton iterations refine poses against both the oracle reference and the independent pilot; all interval weights are then recomputed. Initial rotations have RMS 2° and shifts RMS .5 Å around truth. Search limits are relative to initialization, not truth. This is a controlled local-refinement pipeline against fixed templates, not repeated ab initio density learning.

Each replicate generates alignment image Y_A and independent inference image Y_B . The same fitted design is applied to each, preserving a paired comparison. Nine procedures include folded/sum fixed-pose audits, deterministic pose audit, mixed intervals with shared radii 0/1, and four Gaussian

comparators. The first five retain raw and no-data-fallback results. Each target/procedure has marginal $\alpha = .05$; there is no simultaneous claim over all displayed methods.

The simulation-calibrated joint pose radius is the maximum of 128 dataset-level calibration scores. Under exchangeability with this same generator, template, geometry and optimizer, a new score exceeds it with probability at most $1/129$. Combining this with noise error .04 gives a narrow marginal guarantee for the independent-image deterministic audit. It is known-truth simulation assistance, not experimental pose calibration or a uniform guarantee across new molecules. The mixed-model premise is not asserted for the fitted designs. All 600 datasets finish without exceptions, and all 12,000 interval-weight solves converge. Every one of the 216 procedure/control cells has raw and selected coverage 200/200, with individual exact 95% Monte Carlo interval [.9817, 1]. Consequently, this coarse fixed-generator study does not demonstrate undercoverage of any comparator or a coverage advantage for the proposed bounds.

Table 2 shows the failure of practical informativeness. The three pose-audit procedures always select the no-data interval and never exclude zero. Their raw median widths span 642–1,242 times no data; a post-outcome decomposition attributes 99.31–99.53% of the raw mixed-zero width to its nonlinear remainder. Nevertheless, removing this remainder still leaves all 2,400 first-order bounds wider than no data; the failure is not solely a second-order approximation problem. Fixed-pose and Gaussian intervals have median relative widths .073–.190. They often resolve the central average, whereas contrast sign exclusion is at most .005 for independent inference images. These results do not establish uniform estimated-pose coverage for the simpler procedures.

Median final rotation RMS ranges from 12.0 to 18.5 degrees, compared with the near-truth 2° initialization. The coarse noisy local fit thus worsens poses, even with the oracle template. This is a failure of this declared ten-step local refinement, not evidence about a production pose-refinement pipeline. Pose-ball inclusion ranges from .965 to 1 across realized calibration batches; the rank tolerance guarantee averages over calibration batches as well. The pose bounds here audit the recomputed fixed-audit weights; they are not reoptimized for the pose objective.

6.4. Saved-array diagnosis of alignment bias

After independent review, we declared a post hoc reanalysis of all 600 saved datasets. Every stored estimate replays exactly. Eighteen new deterministic true-pose fits reuse the same observations; all converge. Figure 1 shows audit weights, and all five distinct estimators’ 156 image/control cells are reported in the data. For the pilot-aligned 10049 contrast, independent-image noise-only coverage is .720

Table 2. Local-refinement results: range of median raw half-width/no-data ratios across both targets and templates. All 216 cells have raw and selected coverage 200/200 (individual exact 95% Monte Carlo interval [.9817, 1]). [†]Every trial selects the no-data fallback, with relative width 1 and no sign exclusion. Widths are identical for the paired image controls. This tests the declared recipe, not all possible pose-aware estimators.

Procedure	10028	10049	10076
Fixed folded	0.087–0.107	0.113–0.129	0.106–0.126
Fixed sum	0.089–0.109	0.118–0.133	0.111–0.131
Bounded pose [†]	775–907	987–1039	651–718
Mixed 0 [†]	761–891	973–1026	642–707
Mixed .1 [†]	922–1079	1178–1242	777–856
GP B , fixed	0.136–0.169	0.157–0.190	0.152–0.185
GP B , pose	0.136–0.169	0.158–0.190	0.152–0.185
GP $B/2$, fixed	0.073–0.090	0.088–0.103	0.083–0.101
GP $B/2$, pose	0.073–0.090	0.088–0.104	0.084–0.101

(individual exact 95% interval [.652, .781]), versus .955 with true poses. Mean standardized error rises from .194 to 1.349. The corresponding 10028 central feature has coverage .750 after pilot alignment versus .925 with true poses.

The paired same-minus-independent central estimate shifts by $.3569 \pm .0294$ on 10049 and $.3247 \pm .0245$ on 10076 (mean $\pm$ empirical standard error). Separating each saved error into its signal discrepancy and projected noise shows same-image noise selection: mean projected noise is 1.228 and 1.308 noise SDs for these pilot-aligned features, against .034 and .041 with independent images. Splitting removes this selection effect under the simulation law, while the signal discrepancy remains. RMSE does not uniformly improve on the pilot: for the independent-image 10049 contrast it is 1.350 times the pilot’s absolute error, compared with .999 at true poses.

Reduced-radius folded intervals retain the original weights. Their radii $B \in \{0, .5, 1\}$ all exclude the complete simulated densities: exact pilot-to-generator distances are 1.262, 1.046 and 1.028. Their feature coverage is therefore diagnostic, not a test of an in-class guarantee. The original $B = 2$ intervals remain conservative on these generators.

To diagnose bound slack, we compute the exact continuous class-bias envelope at each realized true/estimated pose pair (Appendix A.2). Across template/target cells its median is .306–.532 of no-data half-width, versus 3.215–5.887 for the first-order interval without its remainder. The old bound is loose, but even the realized oracle envelopes are large. They require true poses and provide neither an implementable interval nor a minimax impossibility result.

6.5. Experimental application and sensitivity

Twelve estimators and 116 files were locked before acquiring 128 new noise-calibration exposures per stack. The inference centers had already been observed. Under a common Gaussian covariance model, the raw calibration ex-

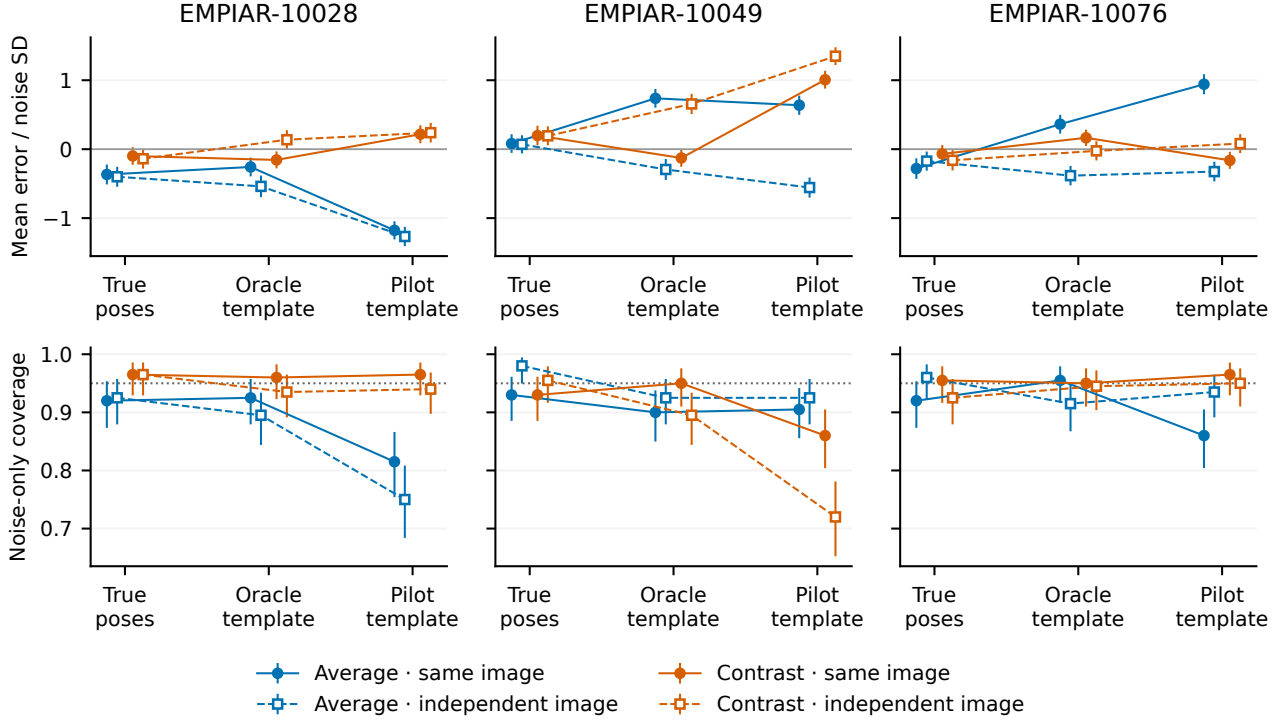


Figure 1. Post hoc replay of all 600 saved datasets, using the audit weights. Top: mean error standardized by each trial’s weight norm, with ± 1.96 empirical standard errors. Bottom: actual coverage of noise-only intervals, with individual exact 95% binomial intervals; dotted line is .95. Both image controls, both targets and all pose controls are shown. True-pose fits are an added control on reused observations. These are narrow noise-only diagnostics, not the original bias-aware intervals, whose coverage remains one.

cludes zero for six fixed-pose features, four with shifts alone, and none with rotation budgets. A later centered calibration changes these counts to ten, six, and zero. It also produces five disagreements with the registered 10076 Class A reference; these are not truth-coverage labels for its heterogeneous particles.

All 96 combinations of feature, pose class, and raw/centered calibration are retained in a fixed-weight breakdown analysis. At the one-degree class, every recorded fixed-weight bias bound already exceeds its observed center even at zero measurement SD. The four centered fixed-pose 10028 density-radius breakdowns span 2.877–4.868 against the declared $B = 2$; two 10049 features do not exclude zero even at $B = 0$. These conditional sensitivities do not validate the scale, density radius, or noise transfer assumptions. More optimized historical cubic designs improve a selected bound modestly; all five experimental cubic designs still include zero.

6.6. Computation and reconstruction context

A separate 10,000-particle quadratic audit completes in 4,965 seconds with 1.95 GB peak resident memory. This establishes that audit’s scaling, not high-resolution recon-

struction speed or convergence of every pose optimizer. Source snapshots, convergence diagnostics, paired replicate arrays, and failed attempts accompany the results.

Supplied-pose Gaussian, voxel and stock-neural fits, three unknown-pose RELION runs, and original-code CryoLike scores provide reconstruction/prediction context. Their FSCs and image scores are not density coverage. The registered-reference mean FSCs for the three RELION runs are .702, .202 and .122; the last uses a heterogeneous stack against one state. Sampled-resolution censoring, map registration, masks and tuning choices are recorded separately (Appendix E). We do not claim performance comparable to a full high-resolution production reconstruction from these restricted studies.

7. Discussion and Limits

The continuous audit tests a representation omission that a restricted model can hide. Its matched Gaussian comparison also shows where an apparent distinction disappears: at equal operator and directional scale, conventional Gaussian uncertainty can share the fixed-pose guarantee. The substantive question is therefore nuisance treatment and the credibility of the assumed class, rather than whether one

estimator is called Bayesian or frequentist.

Estimated alignments change the inference problem. The conditional centered-error theorem requires a model that the fitted designs do not establish; independent inference images alone do not validate it. The end-to-end study measures the resulting procedure and finds that its proposed pose bounds are vacuous. Even the first-order bound is vacuous. Saved-array replay exposes alignment bias and same-image noise selection hidden by the broad class intervals. Realized continuous envelopes show both avoidable bound slack and substantial sensitivity of the declared class. This is a failure of the tested construction, not an impossibility theorem for pose-aware inference. No numerical or empirical result here establishes a useful experimental density interval without the stated unverified noise, pose, support, homogeneity and density-energy conditions.

CTF and amplitude uncertainty, particle selection, motion/dose variation, detector correlations, and learned-prior dependence remain material. Experimental 10076 results are stress evidence, not a homogeneous-density validation. Ordinary floating-point calculations with analytic integration control are not fully validated arithmetic. The work remains a research manuscript; numerical tests and model reviews do not establish conference acceptance.

Reproducibility and Impact

The public repository preserves protocols, code, hashes, complete outcomes, failures, and earlier manuscript versions. The requested independent reviews and responses are available with their unchanged verdicts. Experimental maps should not be interpreted biologically from an uncalibrated interval or internal agreement alone. The purpose of the audit is to make the assumptions and remaining uncertainty inspectable.

References

- Armstrong, T. B. and Kolesár, M. Optimal Inference in a Class of Regression Models. *Econometrica*, 86(2):655–683, 2018. doi: 10.3982/ECTA14434.
- Backus, G. and Gilbert, F. The resolving power of gross earth data. *Geophysical Journal of the Royal Astronomical Society*, 16(2):169–205, 1968. doi: 10.1111/j.1365-246X.1968.tb00216.x.
- Battle, P., Patil, P., Stanley, M., Ruiz Lupon, J., Owahdi, H., and Kuusela, M. Simultaneous Frequentist Calibration of Confidence Regions for Multiple Functionals in Constrained Inverse Problems. *arXiv:2510.11708v1*, 2025. URL <https://arxiv.org/abs/2510.11708v1>.
- Beckers, M., Jakobi, A. J., and Sachse, C. Thresholding of cryo-EM density maps by false discovery rate control. *IUCrJ*, 2019. doi: 10.1107/S2052252518014434.
- Beckers, M., Palmer, C. M., and Sachse, C. Confidence maps: statistical inference of cryo-em maps. *Acta Crystallographica Section D: Structural Biology*, 76:332–339, 2020. doi: 10.1107/S2059798320002995.
- Bharadwaj, A., de Bruin, R., and Jakobi, A. J. Confidence-guided cryo-EM map optimisation with LocScale-2.0. *Nature Communications*, 2026. doi: 10.1038/s41467-026-75327-8.
- Burton Smith, R. N. and Murata, K. Post-acquisition super resolution for cryo-electron microscopy. *IUCrJ*, 13:589–605, 2026. doi: 10.1107/S2052252526005348.
- Chen, M. and Ludtke, S. J. Deep learning-based mixed-dimensional Gaussian mixture model for characterizing variability in cryo-EM. *Nature Methods*, 2021. doi: 10.1038/s41592-021-01220-5.
- Chen, S. and Ling, H. CryoSplat: Gaussian Splatting for Cryo-EM Homogeneous Reconstruction. In *ICLR*, 2026. URL <https://arxiv.org/abs/2508.04929>.
- Chen, S., McMullan, G., Faruqi, A. R., Murshudov, G. N., Short, J. M., Scheres, S. H. W., and Henderson, R. High-resolution noise substitution to measure overfitting and validate resolution in 3D structure determination by single particle electron cryomicroscopy. *Ultramicroscopy*, 135: 24–35, 2013. doi: 10.1016/j.ultramic.2013.06.004.
- Chu, J., Pan, B., and Shi, Z. View-structured conformal prediction for 3d gaussian splatting. *arXiv preprint arXiv:2609.10307*, 2026.
- Chung, S.-C. Cryo-forum: A framework for orientation recovery with uncertainty measure with the application in cryo-EM image analysis. *arXiv:2307.09847v1*, 2023. URL <https://arxiv.org/abs/2307.09847v1>.
- Cossio, P. and Hummer, G. Bayesian analysis of individual electron microscopy images: Towards structures of dynamic and heterogeneous biomolecular assemblies. *Journal of Structural Biology*, 184(3):427–437, 2013. doi: 10.1016/j.jsb.2013.10.006.
- Dai, M., Dong, Z., Xu, K., and Zhang, Q. C. CryoRes: Local Resolution Estimation of Cryo-EM Density Maps by Deep Learning. *Journal of Molecular Biology*, 2023. doi: 10.1016/j.jmb.2023.168059.
- Donoho, D. L. Statistical Estimation and Optimal Recovery. *The Annals of Statistics*, 22(1):238–270, 1994. doi: 10.1214/aos/1176325367.

- Evans, L., Dingeldein, L., Covino, R., Gilles, M. A., Thiede, E., and Cossio, P. Counting particles in cryo-electron microscopy may result in incorrect population estimates. *Communications Biology*, 2026. doi: 10.1038/s42003-026-09859-6.
- Fan, Z., Lederman, R. R., Sun, Y., Wang, T., and Xu, S. Maximum likelihood for high-noise group orbit estimation and single-particle cryo-EM. *The Annals of Statistics*, 52(1):52–77, 2024. doi: 10.1214/23-AOS2292.
- Fiedler, C., Scherer, C. W., and Trimpe, S. Practical and rigorous uncertainty bounds for gaussian process regression. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(8):7439–7447, 2021. doi: 10.1609/aaai.v35i8.16912. Corrected preprint: arXiv:2105.02796v2.
- Galappaththige, C. J., Gottwald, T., Stehr, P., Heinert, E., Suenderhauf, N., Miller, D., and Rottmann, M. Predictive photometric uncertainty in gaussian splatting for novel view synthesis. *arXiv preprint arXiv:2603.22786*, 2026.
- Gilles, M. A. and Singer, A. Cryo-EM heterogeneity analysis using regularized covariance estimation and kernel regression. *Proceedings of the National Academy of Sciences of the United States of America*, 2025. doi: 10.1073/pnas.2419140122.
- Goli, L., Reading, C., Sellán, S., Jacobson, A., and Tagliasacchi, A. Bayes’ rays: Uncertainty quantification for neural radiance fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 20061–20070, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Goli_Bayes_Rays_Uncertainty_Quantification_for_Neural_Radiance_Fields_CVPR_2024_paper.html.
- He, B., Cheng, Y., Li, H., Gao, X., Gao, X., Zhang, F., and Han, R. Resolving compositional and conformational heterogeneity in cryo-EM with deformable 3d gaussian representations, 2026. URL <https://arxiv.org/abs/2512.21599v2>. Version 2, 31 January 2026.
- Hoff, S. E., Thomasen, F. E., Lindorff-Larsen, K., and Bonomi, M. Accurate model and ensemble refinement using cryo-electron microscopy maps and bayesian inference. *PLOS Computational Biology*, 20(7):e1012180, 2024. doi: 10.1371/journal.pcbi.1012180.
- Hoskins, J., Khoo, Y., Mickelin, O., Singer, A., and Wang, Y. Subspace Method of Moments for Ab Initio 3-D Single Particle Cryo-EM Reconstruction. *SIAM Journal on Imaging Sciences*, 19(1):364–409, 2026. doi: 10.1137/24M1699644.
- Janson, A. and Andén, J. Moment-based posterior sampling for multi-reference alignment. *arXiv preprint arXiv:2510.12651*, 2025. URL <https://arxiv.org/abs/2510.12651v1>.
- Jensen, G. J. Alignment error envelopes for single particle analysis. *Journal of Structural Biology*, 2001. doi: 10.1006/jsbi.2001.4334.
- Jeon, M., Raghu, R., Astore, M., Woollard, G., Feathers, R., Kaz, A., Hanson, S. M., Cossio, P., and Zhong, E. D. CryoBench: Diverse and challenging datasets for the heterogeneity problem in cryo-EM. *arXiv:2408.05526v2*, 2025. URL <https://arxiv.org/abs/2408.05526v2>.
- Jia, G., Appia, V., Huang, J., and Wang, X. Rendering-aware bayesian 3d gaussian splatting with native uncertainty and adaptive complexity control. *arXiv preprint arXiv:2607.05522*, 2026.
- Joubert, P. and Habeck, M. Bayesian inference of initial models in cryo-electron microscopy using pseudo-atoms. *Biophysical Journal*, 108(5):1165–1175, 2015. doi: 10.1016/j.bpj.2014.12.054.
- Kanagawa, M., Hennig, P., Sejdinovic, D., and Sriperumbudur, B. K. Gaussian processes and reproducing kernels: Connections and equivalences. *arXiv preprint arXiv:2506.17366*, 2025. URL <https://arxiv.org/abs/2506.17366>.
- Kimanius, D., Jamali, K., Wilkinson, M. E., Lövestam, S., Velazhahan, V., Nakane, T., and Scheres, S. H. W. Data-driven regularization lowers the size barrier of cryo-EM structure determination. *Nature Methods*, 2024. doi: 10.1038/s41592-024-02304-8.
- Lai, T. L., Wang, S.-H., Chung, S.-C., Chang, W.-h., and Tu, I.-P. Uncertainty Quantification in Dynamic Image Reconstruction with Applications to Cryo-EM. *Statistica Sinica*, 33:1771–1788, 2023. doi: 10.5705/ss.202021.0419.
- Levy, A., Raghu, R., Feathers, J. R., Grzadkowski, M., Poitevin, F., Johnston, J. D., Vallese, F., Clarke, O. B., Wetzstein, G., and Zhong, E. D. CryoDRGN-AI: neural ab initio reconstruction of challenging cryo-EM and cryo-ET datasets. *Nature methods*, 2025. doi: 10.1038/s41592-025-02720-4.
- Liao, J., Zheng, D., Zhang, H., Zhang, L., Hu, M., and Bao, C. Fine-tuning AlphaFold with limited cryo-EM observations. *Communications Chemistry*, 9:95, 2026. doi: 10.1038/s42004-026-01899-7.

- Maddipatla, A., Sellam Bojan, N., Bojan, M., Masalitin, V., Vedula, S., Schanda, P., Marx, A., and Bronstein, A. M. Experiment-guided AlphaFold3 resolves measurement-consistent protein ensembles. *Nature biotechnology*, 2026. doi: 10.1038/s41587-026-03166-5.
- Mattingly, H. H., Evans, L., and Cossio, P. Measurement-limited learning of conformational heterogeneity in cryo-electron microscopy. *arXiv:2606.14449v1*, 2026. URL <https://arxiv.org/abs/2606.14449v1>.
- Mordant, G., Evans, L., Silva-Sánchez, D., Cossio, P., and Lederman, R. On the discretization of the object space in inverse problems with application to cryo-electron microscopy, 2026. URL <https://arxiv.org/abs/2609.01688v1>. Version 1, 1 September 2026.
- Narnhofer, D., Habring, A., Holler, M., and Pock, T. Posterior-Variance-Based Error Quantification for Inverse Problems in Imaging. *arXiv:2212.12499*, 2024. URL <https://arxiv.org/abs/2212.12499>. Archived manuscript dated 1 August 2024.
- Nguyen, N. D. and Pham, B. ARCHER: Amortized cross-specimen pose estimation for cryo-electron microscopy. *arXiv:2608.22029v1*, 2026. URL <https://arxiv.org/abs/2608.22029v1>.
- NIST. Digital Library of Mathematical Functions, Section 3.5: Quadrature, 2026. URL <https://dlmf.nist.gov/3.5>.
- Ortiz, S., Stanisic, L., Rodriguez, B. A., Rampp, M., Hummer, G., and Cossio, P. Validation tests for cryo-EM maps using an independent particle set. *Journal of structural biology: X*, 2020. doi: 10.1016/j.jysbx.2020.100032.
- Patel, Y. *Conformally Robust Decision Making*. PhD thesis, University of Michigan, 2025. URL https://www.ambujtewari.com/theses/Yash_Patel_Thesis_2025.pdf. Section 5.2 is a future-direction proposal.
- Punjani, A. and Fleet, D. J. 3DFlex: determining structure and motion of flexible proteins from cryo-EM. *Nature Methods*, 2023. doi: 10.1038/s41592-023-01853-8.
- Qu, H., Wang, X., Zhang, G., Peng, J., and Chen, T. GEM: 3D Gaussian Splatting for Efficient and Accurate Cryo-EM Reconstruction. *arXiv:2509.25075*, 2025. URL <https://arxiv.org/abs/2509.25075>.
- Rangan, A. V., Tang, W.-S., Cossio, P., Zhang, K., and Grigorieff, N. Estimating the tails of the spectrum of the Hessian of the log-likelihood for ab-initio single-particle reconstruction in electron cryomicroscopy. *arXiv:2411.13263v2*, 2024. URL <https://arxiv.org/abs/2411.13263v2>.
- Sanchez Espinosa, D., Thiede, E. H., and Yang, Y. Cryo-EM as a Stochastic Inverse Problem. *arXiv:2509.05541v2*, 2026. URL <https://arxiv.org/abs/2509.05541v2>.
- Sanchez-Garcia, R., Saur, M., Vargas, J., Poelking, C., and Deane, C. M. CESPED: a new benchmark for supervised particle pose estimation in Cryo-EM. *arXiv:2311.06194v2*, 2024. URL <https://arxiv.org/abs/2311.06194v2>.
- Sanchez-Garcia, R., Berndt, A., Apelbaum, A., Reeks, J., Williams, P. A., Poelking, C., Deane, C. M., and Saur, M. Supervised Deep Learning for Efficient Cryo-EM Image Alignment in Drug Discovery with cryoPARES. *bioRxiv*, version 6, 2026. doi: 10.1101/2025.03.04.641536. URL <https://www.biorxiv.org/content/10.1101/2025.03.04.641536v6>.
- Scheres, S. H. W. A Bayesian view on cryo-EM structure determination. *Journal of Molecular Biology*, 415:406–418, 2012. doi: 10.1016/j.jmb.2011.11.010.
- Scheres, S. H. W. and Chen, S. Prevention of overfitting in cryo-EM structure determination. *Nature Methods*, 9: 853–854, 2012. doi: 10.1038/nmeth.2115.
- Schwab, J., Kimanius, D., Burt, A., Dendooven, T., and Scheres, S. H. W. DynaMight: estimating molecular motions with improved reconstruction from cryo-EM images. *Nature Methods*, 2024. doi: 10.1038/s41592-024-02377-5.
- Shabeeb, Z., Saeedi, D., Tsui, D., Jamali, V., and Aghazadeh, A. cryoSENSE: Compressive sensing enables high-throughput microscopy with sparse and generative priors on the protein cryo-EM image manifold. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 34072–34083, 2026. URL https://openaccess.thecvf.com/content/CVPR2026/html/Shabeeb_cryoSENSE_Compressive_Sensing_Enables_High-throughput_Microscopy_with_Sparse_and_Generative_CVPR_2026_paper.html.
- Shaikh, T. R., Hegerl, R., and Frank, J. An approach to examining model dependence in em reconstructions using cross-validation. *Journal of Structural Biology*, 2003. doi: 10.1016/S1047-8477(03)00029-7.
- Sharon, N., Kileel, J., Khoo, Y., Landa, B., and Singer, A. Method of moments for 3D single particle ab initio modeling with non-uniform distribution of viewing angles. *Inverse Problems*, 36(4), 2020. doi: 10.1088/1361-6420/ab6139.

- Shekarforoush, S., Lindell, D. B., Brubaker, M. A., and Fleet, D. J. Reconstructing Heterogeneous Biomolecules via Hierarchical Gaussian Mixtures and Part Discovery. *arXiv:2506.09063*, 2025. URL <https://arxiv.org/abs/2506.09063>.
- Shihab, I. F., Akter, S., and Sharma, A. CalPro: Prior-aware evidential conformal prediction with structure-aware sensitivity bounds for protein structures. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, 2026. URL <https://proceedings.mlr.press/v306/shihab26b.html>.
- Sorzano, C. O. S., Jiménez-Moreno, A., Maluenda, D., Martínez, M., Ramírez-Aportela, E., Krieger, J., Melero, R., Cuervo, A., Conesa, J., Filipovic, J., Conesa, P., del Caño, L., Fonseca, Y. C., Jiménez-de la Morena, J., Losana, P., Sánchez-García, R., Strelak, D., Fernández-Giménez, E., de Isidro-Gómez, F. P., Herreros, D., Vilas, J. L., Marabini, R., and Carazo, J. M. On bias, variance, overfitting, gold standard and consensus in single-particle analysis by cryo-electron microscopy. *Acta Crystallographica Section D Structural Biology*, 2022. doi: 10.1107/S2059798322001978.
- Ullrich, K., van den Berg, R., Brubaker, M., Fleet, D., and Welling, M. Differentiable probabilistic models of scientific imaging with the Fourier slice theorem. In *Proceedings of the 35th Conference on Uncertainty in Artificial Intelligence*, volume 115 of *Proceedings of Machine Learning Research*, 2020. URL <https://proceedings.mlr.press/v115/ullrich20a.html>.
- Van, C. T. S., Reboul, C. F., Caesar, J. J. E., Meana-Pañeda, R., Lontos, G. T., Deme, J. C., Bryant, O. J., Johnson, S., Piczak, C. T., Valkov, E., Lea, S. M., and Elmlund, H. Probabilistic single-particle cryo-EM ab initio 3D reconstruction in SIMPLE. *Acta Crystallographica Section D Structural Biology*, 2025. doi: 10.1107/S2059798325005686.
- Vargas, J., Florea, A. V., Pellegrino, S., Burguet-Castell, J., Gómez-Pedrero, J. A., Arruda Bezerra, G., and Bui, K. H. Gold-standard local validation enhances interpretability of cryo-em maps. Zenodo version 3; attached manuscript filename v5, 17 June 2026, 2026.
- Vilas, J. L., Tagare, H. D., Vargas, J., Carazo, J. M., and Sorzano, C. O. S. Measuring local-directional resolution and local anisotropy in cryo-EM maps. *Nature Communications*, 2020. doi: 10.1038/s41467-019-13742-w.
- Xu, S., Balanov, A., Singer, A., and Bendory, T. Bayesian perspective for orientation determination in cryo-EM with application to structural heterogeneity analysis. *Acta crystallographica. Section D, Structural biology*, 2026. doi: 10.1107/S2059798326001415.
- Yadgar, R. and Lederman, R. R. Best for the eye, not for the algorithm: Anisotropy in fitting atomic models in cryo-em. *bioRxiv*, 2026. doi: 10.64898/2026.07.15.738625. Preprint; primary abstract inspected.
- Yamazaki, K., Wada, M., Wada, Y., Tokuhisa, A., Ohga, K., Nakagawa, A., Akinaga, Y., Sasakura, Y., Katoh, T., Matsumoto, S., and Okuno, Y. Digital twin of biomacromolecular thermodynamics in cryo-EM data. *Communications Chemistry*, 9:293, 2026. doi: 10.1038/s42004-026-02077-5.
- Ye, C. T., Li, S., King, T., and Monakhova, K. QUTCC: Quantile Uncertainty Training and Conformal Calibration for Imaging Inverse Problems. *arXiv:2507.14760v2*, 2026. URL <https://arxiv.org/abs/2507.14760v2>.
- Zhang, A., Mickelin, O., Kileel, J., Verbeke, E. J., Marshall, N. F., Gilles, M. A., and Singer, A. Moment-based metrics for molecules computable from cryogenic electron microscopy images. *Biological Imaging*, 4:e3, 2024. doi: 10.1017/S2633903X24000023.
- Zhang, H., Zheng, D., Wu, Q., Yan, N., Peng, H., Hu, Q., Peng, Y., Yan, Z., Shi, Z., Bao, C., and Hu, M. CryoPROS: Correcting misalignment caused by preferred orientation using AI-generated auxiliary particles. *Nature Communications*, 16:4565, 2025a. doi: 10.1038/s41467-025-59797-w.
- Zhang, J., Zhou, S., Dai, H., Liu, X., Wang, P., Fan, Z., Pei, Y., and Yu, J. CryoFastAR: Fast Cryo-EM Ab initio Reconstruction Made Easy. *arXiv:2506.05864v1*, 2025b. URL <https://arxiv.org/abs/2506.05864v1>.
- Zhao, C., Xu, Y., and Liu, S. When Variance Is Not an Error Map: Calibrated Uncertainty for Radiative Gaussian Splatting in Sparse-View CT. *arXiv preprint arXiv:2607.13682v4*, 2026. URL <https://arxiv.org/abs/2607.13682v4>.
- Zhong, E. D., Bepler, T., Berger, B., and Davis, J. H. Cryo-DRGN: reconstruction of heterogeneous cryo-EM structures using neural networks. *Nature Methods*, 18:176–185, 2021. doi: 10.1038/s41592-020-01049-4.

A. Statistical Identities and Scope

A.1. Normal envelope and continuous Gaussian comparison

Here is a proof of Proposition 1. Write $q(t) = z\sqrt{1+t^2}$ and $F(t) = \Phi(q(t) - t) + \Phi(q(t) + t) - 1$. For $t > 0$, set $c = q'(t) = zt/\sqrt{1+t^2}$. Direct differentiation gives

$$F'(t) = \phi(q - t)\{(c - 1) + (c + 1)e^{-2qt}\}.$$

For $c \geq 1$ this is nonnegative. For $0 < c < 1$, nonnegativity is equivalent to

$$\operatorname{arctanh}(c) \geq qt = \frac{c}{1 - c^2/z^2}.$$

The absolutely convergent power series have coefficients $1/(2j+1)$ and z^{-2j} , respectively. Since $z^2 \geq 3$ and $3^j \geq 2j+1$ for all integers $j \geq 0$, the inequality holds termwise. Thus F is nondecreasing and $F(t) \geq F(0) = 2\Phi(z) - 1$. Symmetry handles negative mean; zero variance follows directly. For positive z the boundary is sharp. Taylor expansion gives

$$F(t) - F(0) = \frac{\phi(z)z(z^2 - 3)}{6}t^4 + O(t^6).$$

When $0 < z < \sqrt{3}$, sufficiently small nonzero bias lowers coverage below its zero-bias level. The case $z = 0$ is the trivial zero-coverage exception.

For $G = \mathcal{A}\mathcal{A}^*$, $a = \mathcal{A}\ell$, $L = \|\ell\|^2$, and fixed positive covariance Σ , completing a quadratic yields

$$v(w) = w^\top \Sigma w + \tau^2 \|\ell - \mathcal{A}^*w\|^2, \quad (13)$$

$$w_* = (G + \Sigma/\tau^2)^{-1}a, \quad (14)$$

$$v(w_*) = \tau^2(L - a^\top w_*). \quad (15)$$

These are the joint Gaussian conditional mean weights and variance for an isonormal prior. With $\Sigma = I$, $\tau = B$, and fixed weights, the worst absolute class bias is $b = B\|\ell - \mathcal{A}^*w\|$. The proposition implies $z\sqrt{v(w)}$ covers every allowed fixed density at level $1 - \alpha$ for $z = z_{1-\alpha/2} \geq \sqrt{3}$. The statement even holds for nonoptimal fixed w if the variance expression on the first line is used. That expression must not be relabeled the exact posterior variance at an unfinished solve. The implication does not require drawing a density from the Gaussian prior. Conversely, posterior calibration averaged over a prior and fixed-density uniform calibration are different statements in general. The matched result here prevents claiming that their distinction alone favors our audit.

An isonormal process $W(f)$ has covariance $E[W(f)W(g)] = \langle f, g \rangle$ for $f, g \in H$. Joint Gaussian laws of $W(\ell)$ and finitely many observation functionals

exist. Identity covariance on infinite-dimensional H has infinite trace, so there is no associated finite-energy random element of H with that covariance. This is a generalized-process comparison, not a claim that all prior draws belong to the bounded density class.

A.2. Exact bias at a realized pose pair

Let A denote the true operator, $\hat{A}$ the fitted rotational operator, and D the orthogonal dephasing by estimated shifts. Write $\delta_0 = w^\top (DA - \hat{A})\rho_0$. For a fixed operator pair and weights,

$$\sup_{\|h\| \leq B} |\delta_0 - \langle \ell - A^*D^*w, h \rangle| = |\delta_0| + B\|\ell - A^*D^*w\|. \quad (16)$$

Cauchy–Schwarz proves the upper bound. When the residual is nonzero, take h with norm B along its direction and opposite the sign of δ_0 ; this attains equality (either sign works if $\delta_0 = 0$). The pose-only analogue replaces the residual by $(A^*D^* - \hat{A}^*)w$. This is elementary Hilbert-space duality applied to the realized operators, not a new coverage principle. Their adjoints are finite Fourier sums, so the sinc identity computes both norms continuously without per-frequency triangle inequalities.

In real/imaginary coordinates, D^*w corresponds to multiplying $w_R + iw_I$ by the conjugate of the dephasing phase. The implementation checks this sign against direct real inner products. Order-40 tensor quadrature integrates the finite sum with an analytic remainder; the remainder for the operator difference uses the union of true and fitted frequencies. Independent dense sinc checks and order-48, tighter-transform-tolerance checks assess numerical error. They do not provide a validated floating-point enclosure.

Equation (16) needs true poses. It is not a computable uniform bound over all unknown poses, and its supremum keeps the weights fixed rather than rerunning alignment for each density. For independent inference noise it describes the conditional signal bias. For same-image alignment it is only the deterministic signal part of the error; projected noise need not be conditionally centered.

A.3. An explicit adaptive-bias counterexample

If $Z \sim N(0, 1)$ and $|\beta(Z)| \leq b$, then $\beta(Z) = b \operatorname{sign}(Z)$ maximizes absolute error pointwise. Thus the worst coverage of a symmetric half-width q is $2\Phi(\max\{q - b, 0\}) - 1$. For $q = c\sqrt{1+b^2}$ and $c > 1$, minimizing over $b \geq 0$ yields

$$\inf_b \Pr\{|Z + \beta(Z)| \leq c\sqrt{1+b^2}\} = 2\Phi(\sqrt{c^2 - 1}) - 1,$$

attained at $b = 1/\sqrt{c^2 - 1}$. The infimum is zero for $0 < c \leq 1$. Consequently an RMS-form bound that covers every adaptive bounded bias at nominal $2\Phi(z) - 1$ needs

$c \geq \sqrt{1+z^2}$. Using $c = z = 1.959964$ instead can reduce coverage to about .908. The ordinary sum bound $b+zs$ is no larger than $\sqrt{1+z^2}\sqrt{s^2+b^2}$ by Cauchy–Schwarz. These are elementary Gaussian calculations, not a new inference principle, and still require a fixed Gaussian noise projection. They do not justify arbitrary noise-dependent weights or estimated variance.

A.4. Optimization, duality, and numerical error

The sum-width problem has dual

$$\min_w z \|w\| + B \|\ell - \mathcal{A}^* w\| = \max_{\|v\| \leq B, \|\mathcal{A}v\| \leq z} \langle \ell, v \rangle. \quad (17)$$

Express the residual norm as the support function of the radius- B ball. Minimizing $z \|w\| - w^\top \mathcal{A}v$ gives zero when $\|\mathcal{A}v\| \leq z$ and minus infinity otherwise. The continuous norm objective and finite observation space satisfy the usual convex duality qualification. A simple feasible certificate is $v = th$ with $h = \ell - \mathcal{A}^* w$ and $t = \min(B/\|h\|, z/\|\mathcal{A}h\|)$, interpreting a zero denominator as no constraint. It gives lower bound $t(L - w^\top \mathcal{A}h)$. We report the gap for this sum objective; selecting a folded-normal width afterwards does not make that width globally optimal.

For $H_\tau = G + \Sigma/\tau^2$ and residual $r = H_\tau w - a$, completing the same quadratic gives

$$v(w) - v(w_*) = \tau^2 r^\top H_\tau^{-1} r. \quad (18)$$

If $FF^\top \preceq G$ is a truncated pivoted-Cholesky approximation and $\Sigma \succeq I$, put $P = FF^\top + I/\tau^2$. Then $P \preceq H_\tau$ and the right-hand side is at most $\tau^2 r^\top P^{-1} r$. Adding a residual diagonal to this preconditioner need not preserve this ordering. If an approximate Gram action has residual error at most e , then

$$\sqrt{r^\top P^{-1} r} \leq \sqrt{r_Q^\top P^{-1} r_Q} + \tau e.$$

We use this as a numerical optimization diagnostic, with the analytic integration remainder below. Roundoff and transform approximation are not validated enclosures, and a small variance gap alone does not bound mean error for every possible observed image vector.

B. Continuous Fourier Computation

B.1. Adjoint and target integrals

Integrating the two conjugate expansions of a real Fourier field yields Equation (7). For a unit-mass Gaussian component of center μ and standard deviation σ_ℓ , its one-dimensional transform truncated to the unit interval is

$$\begin{aligned} & e^{-2\pi i k \mu - 2\pi^2 \sigma_\ell^2 k^2} \\ & \times [\Phi((1/2 - \mu)/\sigma_\ell + 2\pi i \sigma_\ell k) \\ & - \Phi((-1/2 - \mu)/\sigma_\ell + 2\pi i \sigma_\ell k)]. \end{aligned} \quad (19)$$

Products over coordinates and linear combinations over target components give $\mathcal{A}\ell$. Gaussian-product integrals give $\|\ell\|^2$. The sign of the transform follows the convention in the main text. Independent tensor quadrature and a conic formulation using the joint Hilbert Gram of $[\ell, \mathcal{A}_1^*, \dots, \mathcal{A}_m^*]$ check these expressions and the optimizer.

For r Gauss–Legendre nodes on a unit interval, let

$$C_r = \frac{(r!)^4}{(2r+1)((2r)!)^3}, \quad M_a = \max_j |k_{ja}|, \\ E_r = \sum_{a=1}^3 C_r (4\pi M_a)^{2r}.$$

The classical derivative remainder (NIST, 2026) gives, for the positive tensor weights ω_t ,

$$\left| \|f_w\|^2 - \sum_t \omega_t f_w(x_t)^2 \right| \leq E_r \left(\sum_j |c_j| \right)^2. \quad (20)$$

Indeed, the exponential expansion of f_w^2 has coordinate frequencies at most $2M_a$ and coefficient absolute sum at most $(\sum_j |c_j|)^2$. Apply the one-dimensional remainder in each coordinate and telescope the three integration/quadrature operators; their positive weights sum to one. Likewise,

$$\|(G - G_Q)w\| \leq \sqrt{2} E_r \|C/\sigma\|_2 \sum_j |c_j|. \quad (21)$$

We add the scalar error to the squared residual and the vector error to the dual feasibility denominator. This preserves a continuous primal/dual bracket in real arithmetic. At radius five, order 40 gives a kernel remainder no larger than 3.21×10^{-23} before coefficient scaling. Two type-3 transforms apply G_Q . Requested FINUFFT tolerance 10^{-12} is checked against independent sinc Grams; it is not a rigorous floating-point bound. Cached plans change setup overhead, not the mathematical operator.

B.2. Constant-cell generators and spectral Gaussian pilots

An orthonormal constant-cell coefficient has density value $d^{3/2}$ on a cell of volume d^{-3} . Its transform includes the exact cell sinc product $\prod_a \text{sinc}(k_a/d)$ and the half-cell phase relative to a centered FFT grid. Target coefficients use Gaussian CDF differences. Thus a piecewise constant simulation density is projected as a continuous function, not as point samples. Generator grids of 64 and pilot grids of 24 have the same continuous norm convention; unit-normalizing both makes $B = 2$ sufficient by the triangle inequality. It does not calibrate an experimental density-energy radius.

The original fitting representation is a Hermitian sum of Gaussian Fourier kernels. A kernel away from zero fre-

quency has an oscillatory spatial counterpart; a shifted real-space Gaussian instead contributes a phase ramp to a zero-centered Fourier envelope. They should not be conflated. In the present study the Fourier-Gaussian fit supplies a pilot and reconstruction baseline. The continuous audit allows density directions outside that finite representation. Gaussian representations and analytic projection are prior art, not a novelty claim of the uncertainty construction.

C. Pose Derivatives, Remainders, and Calibration

C.1. First derivative and second-order control

For each particle write $c_{iq} = (w_{R,iq} + iw_{I,iq})C_{iq}/\sigma$. Use a rotation scale a in radians, detector translation scale s_t in field units, and dimensionless local error $u \in \mathbb{R}^5$. Its rotation is $\exp(a[u_{1:3}]_\times)$ and translation $s_t u_{4:5}$. The five first-adjoint derivatives are real polynomial Fourier fields: rotation columns have polynomial $2\pi i a c_{iq}(k_{iq} \times e_j)^\top x$; translation columns have $2\pi i s_t c_{iq} q_j$. The sign convention matches the implemented dephasing and is checked by finite differences.

Their 5×5 Hilbert Gram follows from integrals of $[1, x, y, z][1, x, y, z]^\top e^{2\pi i v^\top x}$ at both frequency sums and differences. These integrals are derivatives of $K(v)$. The square root of the largest Gram eigenvalue gives $\|D_i\|_{\text{op}}$. A small cancellation pad is a floating-point diagnostic, not a proved rounding bound. The pilot derivative pairing is integrated using the same exact constant-cell representation.

For a unit vector b and x uniform on the cube, $E(b^\top x)^2 = 1/12$ and $E(b^\top x)^4 \leq 13/720$. Along a unit pose segment the phase derivative consequently satisfies

$$\|\phi'\|_4 \leq L_{4,iq} = 2\pi\{a\|k_{iq}\|(13/720)^{1/4} + s_t\|q_{iq}\|\}, \quad (22)$$

$$\|\phi''\|_2 \leq H_{2,iq} = 2\pi a^2\|k_{iq}\|/\sqrt{12}. \quad (23)$$

Since $(e^{i\phi})'' = e^{i\phi}(i\phi'' - (\phi')^2)$, Minkowski and Hölder give a second derivative norm at most $H_{2,iq} + L_{4,iq}^2$. Writing $P_0 = \|\rho_0\|$, Taylor's integral remainder and Cauchy-Schwarz yield the uniform scalar bound

$$R_2 = \frac{B + P_0}{2} \sum_i (r_i + \eta_i)^2 \sum_q |c_{iq}|(L_{4,iq}^2 + H_{2,iq}). \quad (24)$$

This bound remains valid for large prescribed radii but can become useless. It retains all second-order and higher effects through the exact derivative bound along the segment; it is not a second-order truncation without a remainder. Triangle bounds lose possible cancellation and are not claimed sharp. The mixed-pose theorem follows by combining this remainder with Hoeffding's lemma as in the main text.

The historical quadratic/cubic audits retain shared density fields across particles and add higher-order integration-controlled remainders. Their complete code, proofs, failed optimizers, and outcomes remain in the repository. They are not substituted into the frozen local-refinement experiment after seeing its results.

C.2. Simulation-assisted maximum-score calibration

Let $S_1, \dots, S_m, S_{m+1}$ be exchangeable dataset scores for one fixed truth, geometry, optimizer, and template. Rank symmetry, with random tie-breaking if needed, gives

$$\Pr\{S_{m+1} > \max_{j \leq m} S_j\} \leq 1/(m+1).$$

The score here is the maximum over particles of $\sqrt{(\text{angle in degrees})^2 + (\text{shift in } \text{\AA}/0.5)^2}$. For $m = 128$, its maximum determines a joint rotation/shift ball for the next dataset. On that event, the deterministic bound with independent inference images has conditional noise error at most .04. Union with the score failure yields unconditional error at most $.04 + 1/129 < .05$. This averages over calibration as well as test datasets. It is not a conditional guarantee for every realized calibration batch. The pilot and truth are fixed; it is not distribution-free across new molecules or an experimentally learned pose-confidence ball. Same-image inference lacks the required conditional noise law.

D. Phase Alignment Diagnostic

D.1. Bias after same-image and independent-image alignment

Let $A_j = se^{i\theta_j} + \sigma(Z_{1j} + iZ_{2j})$ and $B_j = se^{i\theta_j} + \sigma(W_{1j} + iW_{2j})$, with all real coordinates independent standard normal, fixed $s > 0$, and arbitrary fixed true phases. Set $\hat{\theta}_j = \arg A_j$. Rotational symmetry reduces the mean calculation to $\theta_j = 0$. Strict Jensen inequality for the norm gives $E|A_j| > s$. Independence gives $E \operatorname{Re}(e^{-i\hat{\theta}_j} B_j) = sE \cos \hat{\theta}_j < s$ when $\sigma > 0$. Both errors persist as more particles are independently aligned. The law of large numbers and any half-width tending to zero imply coverage of s tends to zero. Estimating the aligned observations' sampling variance does not remove their bias. This elementary control diagnoses the mechanism; it is not an impossibility theorem for marginalized or invariant inference.

D.2. Invariant comparison with a finite-sample bound

Put $X_j = \operatorname{Re}(\bar{A}_j B_j)$ and $m = s^2$. Then $EX_j = m$ and $\operatorname{Var}(X_j) = 2(\sigma^4 + m\sigma^2)$. A product-normal Gaussian integral, or the difference of independent noncentral and

central chi-squares, gives for $|t| < 1/\sigma^2$

$$\log Ee^{t(X_j - m)} = -\log(1 - \sigma^4 t^2) + \frac{m\sigma^2 t^2}{1 - \sigma^2 t}. \quad (25)$$

For verification, take real two-vectors $U = (A + B)/(\sqrt{2}\sigma)$ and $V = (A - B)/(\sqrt{2}\sigma)$; their squared norms have two-degree chi-square laws, with noncentrality $2m/\sigma^2$ for U . Since $X = (\sigma^2/2)(\|U\|^2 - \|V\|^2)$, the displayed expression follows. For either sign of t , the centered log MGF is bounded by $(\sigma^4 + m\sigma^2)t^2/(1 - \sigma^2|t|)$. The independent-sum Chernoff bound with $x = \log(2/\alpha)$ implies

$$\Pr \left\{ |\bar{X} - m| > 2\sqrt{(\sigma^4 + m\sigma^2)x/n} + \sigma^2 x/n \right\} \leq \alpha. \quad (26)$$

Invert over $m \geq 0$ and square-root endpoints to obtain a confidence set for s . Empty sets are saved as empty, not converted to $[0, 0]$. The guarantee requires known equal Gaussian coordinate noise and identical signal across the pair; motion, dose, CTF error and heterogeneous amplitudes would require a different analysis. This is an elementary concentration specialization and a classical invariant control, not new cryo-EM theory.

E. Experimental Protocols and Complete Reporting

E.1. Data provenance and independence

The project downloads experimental particles and metadata from EMPIAR and reference maps from EMDB. Exact URLs, selected indices, file hashes, signs, physical fields and software versions accompany each artifact. EMPIAR-10028, 10049 and 10076 supply three distinct acquisition geometries, not three independent repetitions of a molecular population. Reference maps are approximate external structures used as simulation generators and agreement comparators. Pilot-only frame/hand registration is frozen before the registered replay; the same maps in their original frame remain reported. Registration does not turn a map into experimental ground truth.

Particle splits use acquisition groups. Within the existing particle cohorts, poses and CTF metadata may derive from full-data processing; a particle split does not undo this dependence. The noise-calibration experiment acquires 128 additional exposures per stack after locking the estimators and 116 calibration-related files. Its density interval centers had already been inspected, and the common covariance and transfer assumptions remain unverified. Selection of the three pilot regions and matched center precedes those fresh calibration images, but the full project is development research rather than a prospectively blinded experimental confirmation.

EMPIAR-10076 contains assembly intermediates. A homo-

Table 3. All registered dictionary replays. Coverage is analytic under the supplied fixed design/noise. Ambient-audited coverage is numerically one in both frames for these generators; this is not experimental density coverage.

Stack	Target	SD/field	Original	Registered
10028	Center	0.03	0	0
10028	Contrast	0.03	0.916	0.054
10028	Center	0.07	4.18×10^{-5}	3.73×10^{-14}
10028	Contrast	0.07	0.800	0.367
10049	Center	0.03	0.814	0.759
10049	Contrast	0.03	0.002	0.507
10049	Center	0.07	0.871	0.928
10049	Contrast	0.07	4.90×10^{-16}	0.786
10076	Center	0.03	0.001	0.250
10076	Contrast	0.03	0.605	0.380
10076	Center	0.07	0.918	0.942
10076	Contrast	0.07	0.670	0.899

geneous simulation using registered EMD-8434 Class A is a legitimate fixed-generator test but is not ground truth for its heterogeneous experimental stack. All biological interpretations of those experimental consensus features retain this limit.

E.2. Registered representation replay

The twelve original dictionary examples use two targets and two widths on each geometry. Existing weights exactly replay their archived widths, biases and analytic coverages before the reference frame is changed; the maximum stored discrepancy is zero. Native versus two-stage resampling changes the original generator’s projections by at most 2.15×10^{-7} relatively. There is no refit to favor the registered comparison. The finite ambient audit and the continuous audit concern different declared spaces. Both illustrate representation sensitivity, but an ambient-voxel result is not relabeled a continuous theorem. Near-unit coverage on a particular generator can coexist with wide or biologically unhelpful intervals.

E.3. Matched continuous Gaussian comparison

The frozen comparison has 128 particles per geometry, frequency radius 12, geometry seed 609315, order-80 quadrature, and 20 Å targets. It uses the same twelve pilot-locked locations as the continuous audit, $B = 2$, prior scales 2 and 1, and $\alpha = .05/12$. Fixed-pose covariance is identity after supplied whitening. The local-Gaussian alternative uses the actual continuous pilot Jacobian, with coordinate SDs 1° and $.5 \text{ Å}$. This is a first-order working model, not the bounded nonlinear pose class.

For each of 48 fits, both original and registered generators are projected at nominal poses and at all six declared perturbations: rotation scales 0, 1 and 2 degrees, each with coherent and random-boundary directions and translation

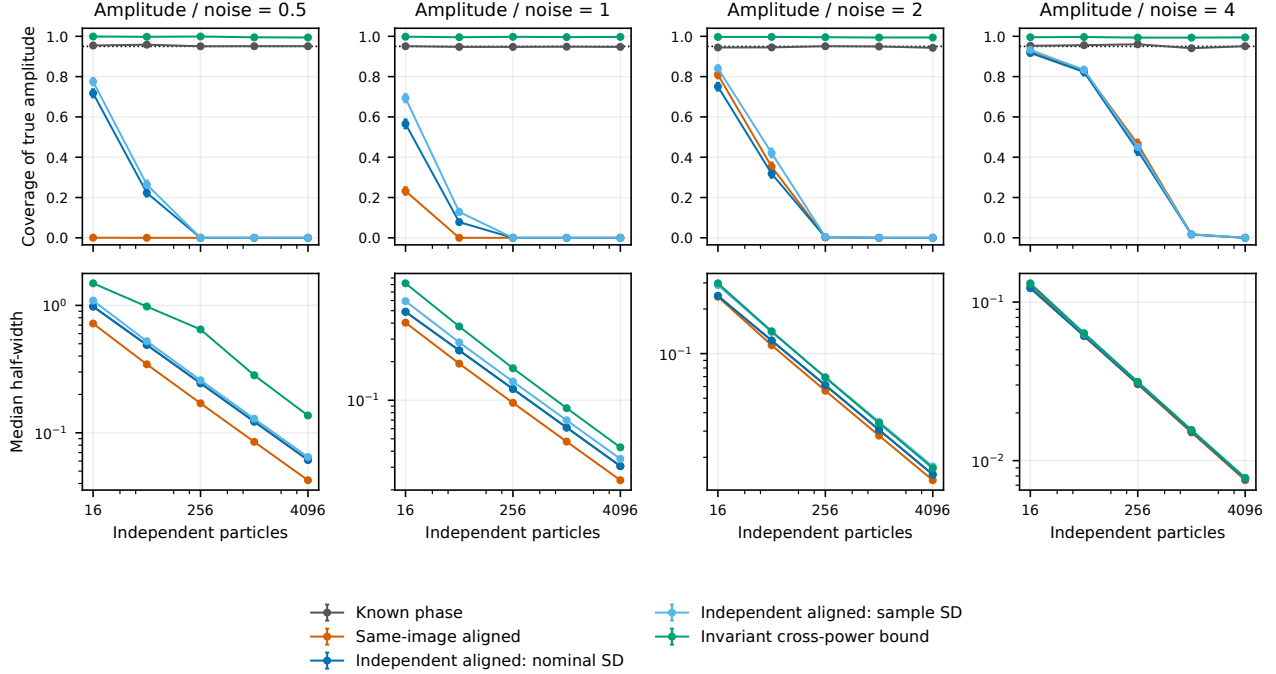


Figure 2. Declared phase-only control: 2,000 independent datasets per case, five paired procedures, four amplitude-to-coordinate-noise ratios, and five particle counts. At 4,096 particles all three naive aligned intervals have zero observed coverage in every noise setting. Independent-image sample variance does not correct attenuation. Oracle coverage is near .95; the invariant concentration interval is conservative. Width and coverage are both retained. This is a one-frequency mechanism diagnostic with known noise, not a reconstruction benchmark.

scale $.5 \text{ \AA}$. The zero-degree perturbation setting can still include shifts and differs from the nominal case. Conditional fixed-signal coverage and correct-sign probability are computed from their exact scalar Gaussian laws under the supplied noise. These are not end-to-end coverage experiments. All 672 fit/frame/scenario evaluations are retained.

The initial rank-1,024 implementation was interrupted after four fits, all nonconverged at the 2,000-iteration limit. Its first residual was about 1.04×10^{-5} and a residual-based variance-gap diagnostic was about 9%. Its JSON, weights, logs and incomplete flag are preserved. A numerical pre-flight, without new pixels or coverage selection, increased the preconditioner rank to 8,192: after 315 seconds of setup, the first system reached residual 9.50×10^{-12} in three warm-started iterations. The declared complete rerun uses this rank, cached transforms, zero initial weights, one CPU thread, the same 10^{-10} tolerance and 2,000-iteration ceiling, and unchanged scientific settings. All convergence flags and variance-identity errors remain visible. This amendment is computational, not a replacement of an unfavorable statistical baseline. The three complete setup-and-fit runs take 518, 384 and 373 seconds, respectively, on the authorized Mac under concurrent load. The process peak resident memory is 6.08 GB; this is a cumulative process high-water mark, not a fresh per-geometry memory measurement. All relative CG

Table 4. Minimum fixed-pose uniform class coverage across four targets, at nominal $1 - .05/12 = .995833$. Columns separate the prior scale and fixed/linearized covariance weights. These are analytic class minima, not coverage on a selected generator.

Stack	2, fixed	2, linear	1, fixed	1, linear
10028	0.999930	0.999931	0.979915	0.980143
10049	0.995836	0.995842	0.994882	0.994891
10076	0.995840	0.995851	0.995363	0.995375

residuals are at most 9.61×10^{-11} . The original rank-1,024 failures are separate from these 48 successful numerical fits. At prior scale $B/2$, class-worst-case coverage falls below nominal on all three stacks, although coverage of the selected simulation generators is favorable. At scale B , all 24 class minima exceed .995836. This distinction prevents treating coverage of benign generators as a uniform class result.

E.4. Local-pose refitting study

The calibration seed root is 261001; the test root is 261002. Each is combined with dataset identifier and replicate index using NumPy’s seed sequence. Calibration contains 128 datasets per geometry; testing has exactly 200. New alignment noise, inference noise and local initialization are

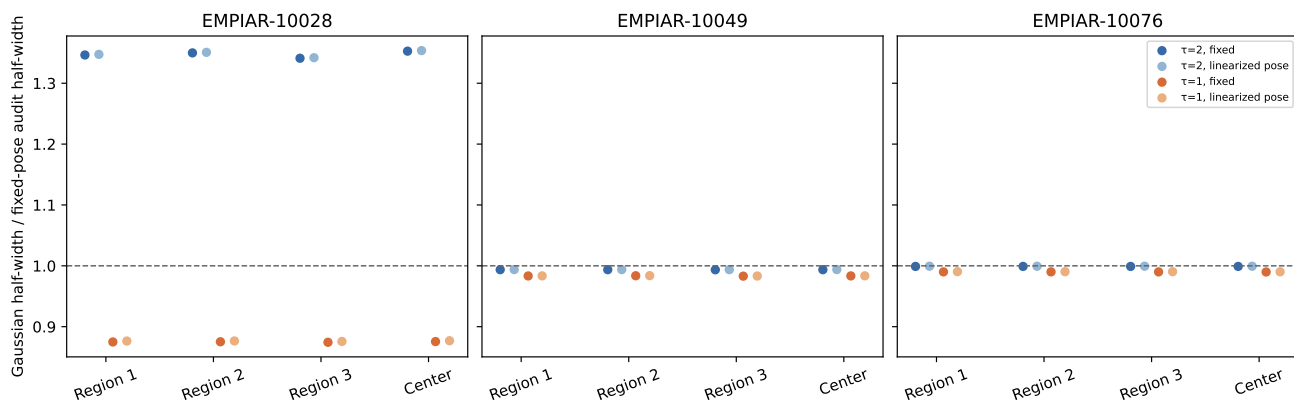


Figure 3. All twelve locked features and four continuous Gaussian comparisons. The dashed line denotes the original fixed-pose audit width. Matched prior scale $\tau = 2$ gives near-equal widths on two stacks; its 10028 widths are 34–35% larger. The half-scale prior is narrower without the same uniform class implication. Adding the specified first-order Gaussian pose covariance has little effect here. Every fit converged.

independently drawn for each replicate. Templates, truth and acquisition geometry remain fixed. All templates, targets, methods and image controls within that replicate are paired, not independent observations.

The optimizer uses the frozen ten-iteration Gauss–Newton procedure and its unchanged step/search constraints. Oracle-template and independent-pilot fits share the noisy alignment images and initialization. Pose search is local around near-truth initialization; we do not claim global orientation discovery. Estimated translations dephase both observation controls and estimated rotations define a new Fourier Gram. Every interval weight is recomputed. The affine inference pilot remains the independent pilot even when the alignment template is oracle truth. Thus the oracle control isolates reference quality for alignment, not an oracle density estimator.

Fixed-audit weights minimize the sum objective with 100 maximum iterations and relative-gap target .005. Gaussian fits use relative CG tolerance 10^{-10} and at most 2,000 iterations. Quadrature order is 40 and preconditioner rank 1,024. Local-Gaussian pose SDs are calibration RMS errors per coordinate, without removing mean error. These differ from the fixed $1^\circ/5 \text{ \AA}$ values in the separate radius-12 comparison.

All nine procedures, both targets, both templates and both image controls give 72 interval cells per replicate. The first five procedures retain raw and no-data-selected results; all four Gaussian procedures are reported raw. Selection compares widths using the fitted design, not the inference value. When the no-data interval wins, both center and width switch to the pilot/class interval. It can obtain coverage with no sign information, so fallback rate and raw outcomes must accompany selected coverage.

Every attempted replicate writes a JSON and arrays, including partial arrays and exception text on failure. Failed trials

remain in the planned denominator of 200. A completed replicate must contain all 72 unique cells. Summaries require all 200 attempts and recheck hashes of every JSON, array and calibration input. Exact binomial 95% intervals quantify Monte Carlo uncertainty separately for each procedure; they are not simultaneous confidence bands over all displayed comparisons. Width summaries use the available outcomes and report their count. Paired image-control discordance counts and center differences are retained. Empirical cross-particle correlations and mean errors are unconditional diagnostics, not tests that establish conditional-on-design centering. All 600 datasets and 12,000 solves complete. The total recorded test runtime is 19,584 seconds; peak resident memory is below .681 GB per process. All 216 raw/selected coverage fractions are one. The selected pose intervals use no inference information, so their perfect observed coverage is not an informativeness result. All paired coverage discordance counts are zero. The no-data reference is itself conditional on the supplied pilot-centered class; relative width is not biological resolution.

The final pose errors are substantially larger than the near-truth initialization: rotation RMS medians span 11.990–18.481 degrees and shift medians 6.617–9.282 $\text{\AA}$. Calibration radii are determined by these same poor local-fitting dynamics. The post-outcome arithmetic remainder decomposition retains all 2,400 template/target fits and does not change an interval or scientific parameter. Minimum and maximum nonlinear-remainder fractions of the raw mixed-zero width are .993056 and .995307. The detailed records include all pose errors, unconditional dependence diagnostics, centers, absolute widths, widths relative to estimates and paired differences. All 216 detailed coverage, fallback and sign-exclusion rows and both heat maps remain in the machine-readable release. Since every coverage entry is one and every pose fallback entry is one, the compact main

table reports their common outcome without repeating three pages of identical entries.

E.5. Experimental breakdown analysis

For each fixed estimator, the stored pilot, density-radius coefficient, pose remainder and noise scale determine a width as B or measurement SD is varied. The reporting script reconstructs the stored $B = 2$ result to 10^{-9} tolerance before solving where the interval first contains zero. It reports all 96 feature/pose/calibration combinations; it does not refit weights at each radius. A breakdown of zero means the feature already fails sign exclusion at zero radius or noise. It is not evidence that the true density radius is zero.

At one degree, every retained fixed-weight bias bound already exceeds the observed center at zero measurement SD. At fixed poses, centered-calibration 10028 breakdowns range from 2.877 to 4.868. These sensitivities measure how far an assumed input can move before a reported conclusion changes; they do not estimate that input or validate the common-noise law. The inference centers were known before this analysis. All raw/selected widths, no-data-fallback flags and registered-reference disagreements remain saved.

E.6. Reconstruction baselines and FSC interpretation

Stock cryoDRGN 4.3.1 homogeneous training uses supplied consensus poses, three hidden layers of width 256 and batch size eight, separately on both exposure-group halves of each stack. These are actual neural fits, not backprojection, but are not *ab initio* CryoDRGN-AI or uncertainty inference. Gaussian and voxel ridge fits use those same halves. Held-out image scoring uses identical particles, 1,410 Fourier pairs per image inside radius 30 of a 64-pixel grid, and the same window/background normalization. Stock neural training uses its radius-32 loss and per-half global scale, so the training objectives differ. Tuning-group prediction error selects epochs 60, 50 and 100; two are final available checkpoints and undertraining remains possible. The development stopping budgets are not preregistered convergence criteria.

RELION 5.0.1 learns poses on all three stacks using a common prescribed initialization/refinement protocol, with exposure-half labels verified. Wall-limited initial attempts and separately declared continuations remain available. No checkpoint is chosen by reference FSC. Native half FSC is computed before map registration, unmasked, with the coupled low-frequency region displayed: RELION joins halves below $1/40 \text{ \AA}^{-1}$. Cross-method and external-reference FSC use the fixed pilot-only frame/hand transform, retaining interpolation and registration limits. All native and registered curves are provided, including poor agreement.

For 10049, native FSC crosses .5 at 22.73 $\text{\AA}$ and .143 at

18.95 $\text{\AA}$. For 10076 these crossings are 24.73 and 15.92 $\text{\AA}$. The registered-reference mean FSCs are .702, .202 and .122 for 10028, 10049 and 10076. Mean FSC is a descriptive average over the recorded shells, not a resolution measure. The heterogeneous third stack and state-specific reference limit interpretation. Half-map agreement above a threshold through the sampled band is reported as censored at that limit, not an observed threshold crossing. Algorithmic convergence also does not establish structural accuracy.

Original-code CryoLike likelihood scoring and held-out prediction provide additional image-fit context. A likelihood or FSC is not a confidence interval for the unknown true feature; we do not use those methods as if they promised such coverage. Conversely, weak reconstruction agreement is not hidden by reporting only an uncertainty diagnostic. Figures 5–7 show every method in the same three central planes, with display normalization stated explicitly.

E.7. Raw-movie feasibility and remaining dependence

A separately declared acquisition pilot downloaded one EMPIAR-10028 movie, `Micrographs_part2/004_movie.mrcs`, with 16 frames of 4096×4096 float32 pixels. The source-group mapping is supported by archive counts, acquisition date and nearby coordinates, but the deposited autopicks differ from final recentered coordinates. It is not labeled a fully verified new confirmatory cohort. Its completed analysis is restricted to all-frame acquisition statistics, a fixed patch grid, frame-difference spectra/correlations and unaligned odd/even sums. Differences can contain motion, dose signal and detector effects; they are not automatically pure noise or proof of frame independence. No density intervals result from this pilot. Any subsequent motion, picking, CTF or alignment pipeline requires its own declared processing graph and cannot claim independent inference noise while silently reusing full-movie-derived quantities.

All sixteen frames, 512 fixed patch differences and 1,792 cross-pair correlations were retained. Median horizontal and vertical neighbor correlations are .657 and .658; the 5th–95th percentiles of disjoint difference correlation are $-.00780$ to $.00866$. Frame means span 2052.53–2060.28 in deposited pixel units. The difference fields are spatially correlated; these statistics do not separate detector noise from residual signal. Small temporal correlations do not establish independence or a transferable covariance. The interrupted download and verified byte-range recovery are preserved; they did not change the predeclared movie or diagnostics.

Table 5. Post hoc audit-weight diagnostics. Each pair is same-image / independent-image; every cell has 200 paired trials. RMSE is divided by the fixed pilot absolute error. The complete release includes all five distinct estimators, all radius choices, and individual Monte Carlo intervals.

Stack	Pose control	Target	Mean error / SD	Noise-only coverage	RMSE / pilot error
10028	True poses	center	-0.367 / -0.400	0.920 / 0.925	0.061 / 0.062
10028	True poses	contrast	-0.099 / -0.139	0.965 / 0.965	0.276 / 0.311
10028	Oracle template	center	-0.258 / -0.539	0.925 / 0.895	0.056 / 0.069
10028	Oracle template	contrast	-0.156 / 0.137	0.960 / 0.935	0.265 / 0.297
10028	Pilot template	center	-1.177 / -1.266	0.815 / 0.750	0.085 / 0.092
10028	Pilot template	contrast	0.217 / 0.239	0.965 / 0.940	0.299 / 0.325
10049	True poses	center	0.081 / 0.070	0.930 / 0.980	1.432 / 1.391
10049	True poses	contrast	0.198 / 0.194	0.930 / 0.955	1.029 / 0.999
10049	Oracle template	center	0.737 / -0.295	0.900 / 0.925	1.590 / 1.478
10049	Oracle template	contrast	-0.128 / 0.656	0.950 / 0.895	0.757 / 1.018
10049	Pilot template	center	0.638 / -0.557	0.905 / 0.925	1.600 / 1.594
10049	Pilot template	contrast	1.007 / 1.349	0.860 / 0.720	1.127 / 1.350
10076	True poses	center	-0.282 / -0.173	0.920 / 0.960	0.343 / 0.314
10076	True poses	contrast	-0.068 / -0.161	0.955 / 0.925	0.758 / 0.872
10076	Oracle template	center	0.363 / -0.384	0.955 / 0.915	0.317 / 0.330
10076	Oracle template	contrast	0.165 / -0.024	0.950 / 0.945	0.689 / 0.797
10076	Pilot template	center	0.941 / -0.325	0.860 / 0.935	0.433 / 0.332
10076	Pilot template	contrast	-0.162 / 0.081	0.965 / 0.950	0.718 / 0.792

Table 6. Density-radius breakdowns for every locked feature and both experimental noise calibrations. Each pair is raw/centered calibration. Zero means sign exclusion already fails at $B = 0$; the declared radius is $B = 2$. These are fixed-weight sensitivity thresholds, not estimates of the true radius. Every one-degree noise-scale breakdown is zero. Full half-width/center ratios, noise thresholds and fallback flags are retained in the all-case CSV.

Stack	Feature	Fixed poses	Shifts only	One degree	Two degrees
10028	Region 1	3.447/4.832	0.516/0.757	0.000/0.000	0.000/0.000
10028	Region 2	3.444/4.868	0.542/0.807	0.000/0.000	0.000/0.000
10028	Region 3	1.595/2.877	0.187/0.395	0.000/0.000	0.000/0.000
10028	Center	1.612/3.055	0.193/0.457	0.000/0.000	0.000/0.000
10049	Region 1	0.000/153.172	0.000/3.401	0.000/0.000	0.000/0.000
10049	Region 2	0.000/0.000	0.000/0.000	0.000/0.000	0.000/0.000
10049	Region 3	0.000/0.000	0.000/0.000	0.000/0.000	0.000/0.000
10049	Center	0.000/133.654	0.000/2.854	0.000/0.000	0.000/0.000
10076	Region 1	119.076/186.381	7.143/11.239	0.000/0.054	0.000/0.000
10076	Region 2	68.808/126.048	4.158/7.692	0.000/0.000	0.000/0.000
10076	Region 3	111.064/178.140	6.626/10.713	0.000/0.016	0.000/0.000
10076	Center	89.151/161.521	5.332/9.752	0.000/0.000	0.000/0.000

Table 7. Frozen additional-exposure prediction: 4,096 particles per dataset from 115, 71 and 185 source groups absent from development. Models and cohort were locked before image access. Lower normalized prediction MSE is better. Supplied consensus poses/CTFs remain conditional; this does not measure unknown density coverage.

EMPIAR	Gaussian	Voxel	Neural
10028	0.835319	0.841379	0.829369
10049	0.931579	0.932359	0.930739
10076	0.874799	0.879078	0.868657

EMPIAR	Comparator minus neural	NMSE difference	Adjusted percentile interval
10028	Gaussian	0.005950	[0.004995, 0.006932]
10028	Voxel	0.012011	[0.010695, 0.013356]
10049	Gaussian	0.000840	[0.000659, 0.001023]
10049	Voxel	0.001620	[0.001366, 0.001875]
10076	Gaussian	0.006143	[0.005686, 0.006593]
10076	Voxel	0.010421	[0.009792, 0.011038]

All six prespecified paired differences are shown. Intervals use 10,000 whole-exposure bootstrap resamples and a Bonferroni adjustment for six contrasts (99.1667% marginal percentile intervals). This is a bootstrap approximation, not a finite-sample coverage theorem. No model selection uses these outcomes.

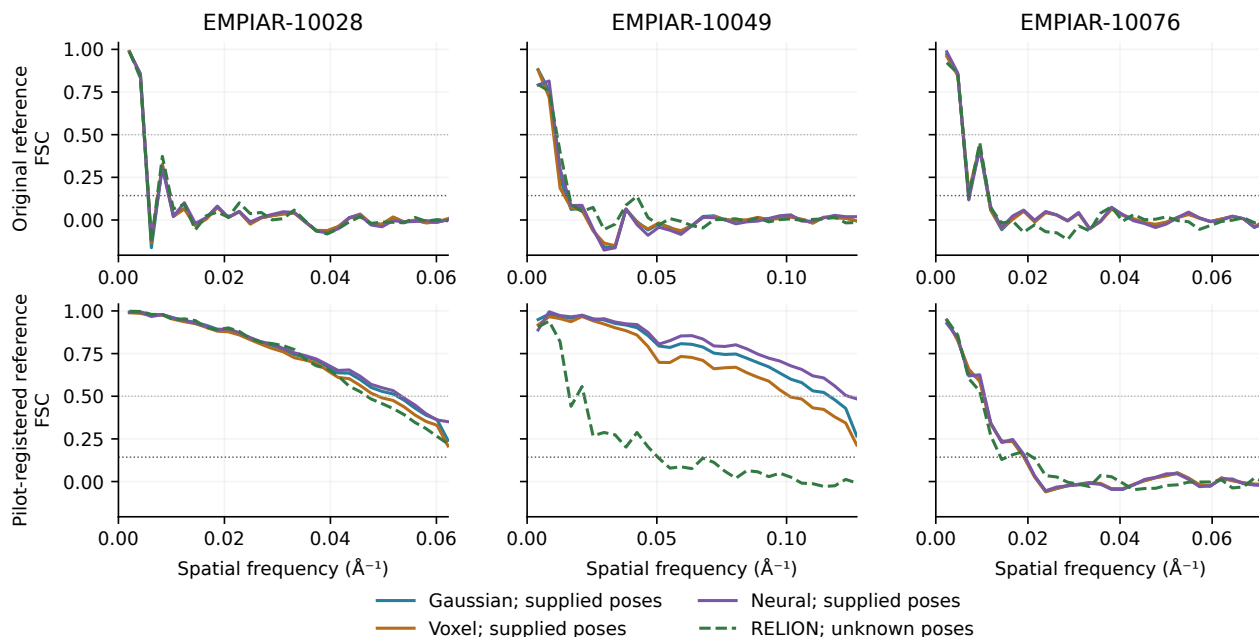


Figure 4. All twelve reconstruction method/stack comparisons against original and pilot-registered references; dotted lines mark FSC 0.5 and 0.143. Gaussian, voxel and neural fits use supplied poses; RELION estimates poses. The old pilot alone determines registration. Agreement includes preprocessing, interpolation and frame dependence. Censored crossings are not measured resolutions. The 10076 Class A reference is not truth for its heterogeneous consensus, and none of these curves measures interval coverage.

Table 8. Predeclared acquisition diagnostics from one raw 10028 movie. Difference statistics use every one of 512 patch/pair combinations; disjoint-pair correlations use all 1,792 comparisons. Quantiles are descriptive across dependent summaries, not independent-replicate confidence intervals.

Statistic	Minimum	5th percentile	Median	95th percentile	Maximum
Difference SD	341.7	351.2	357.7	370.3	402.5
Horizontal neighbor correlation	0.6455	0.6496	0.6569	0.6641	0.6844
Vertical neighbor correlation	0.6463	0.6522	0.6583	0.6662	0.6862
Disjoint-difference correlation	-0.01856	-0.007798	0.0006576	0.008661	0.01639

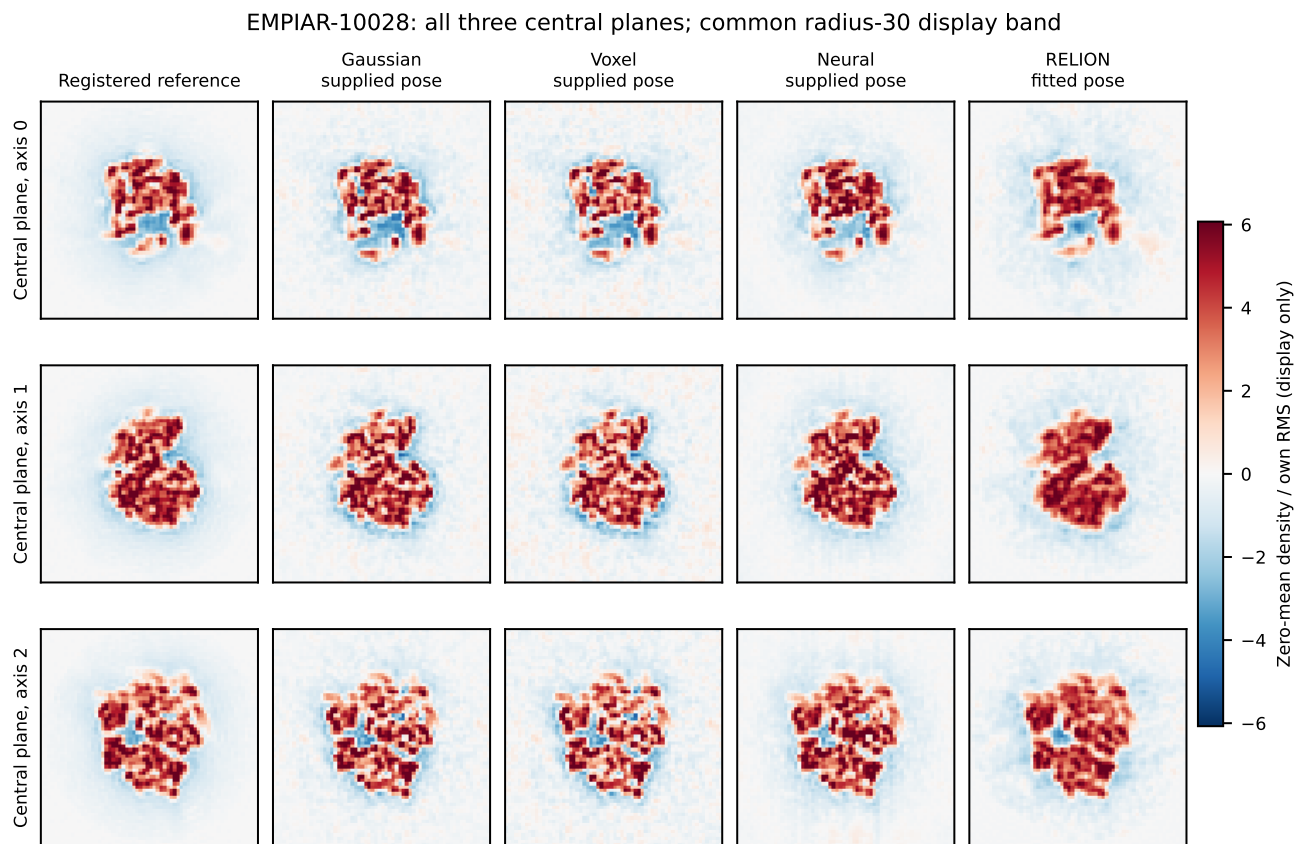


Figure 5. EMPIAR-10028: all three central planes of the registered reference and four existing reconstructions. Each square spans $482.4 \text{ \AA}$. The display removes DC, retains a common radius-30 Fourier band, and divides each map by its own volumetric RMS; a pooled 99.5th-percentile absolute value fixes the common symmetric color range. This display normalization hides amplitude differences and is not used for FSC or intervals. Views are fixed array planes, not selected for agreement. Gaussian, voxel and neural estimates average the locked halves; RELION uses its previously evaluated map. Registration is pilot-derived. The 10076 reference represents Class A, not the heterogeneous consensus.

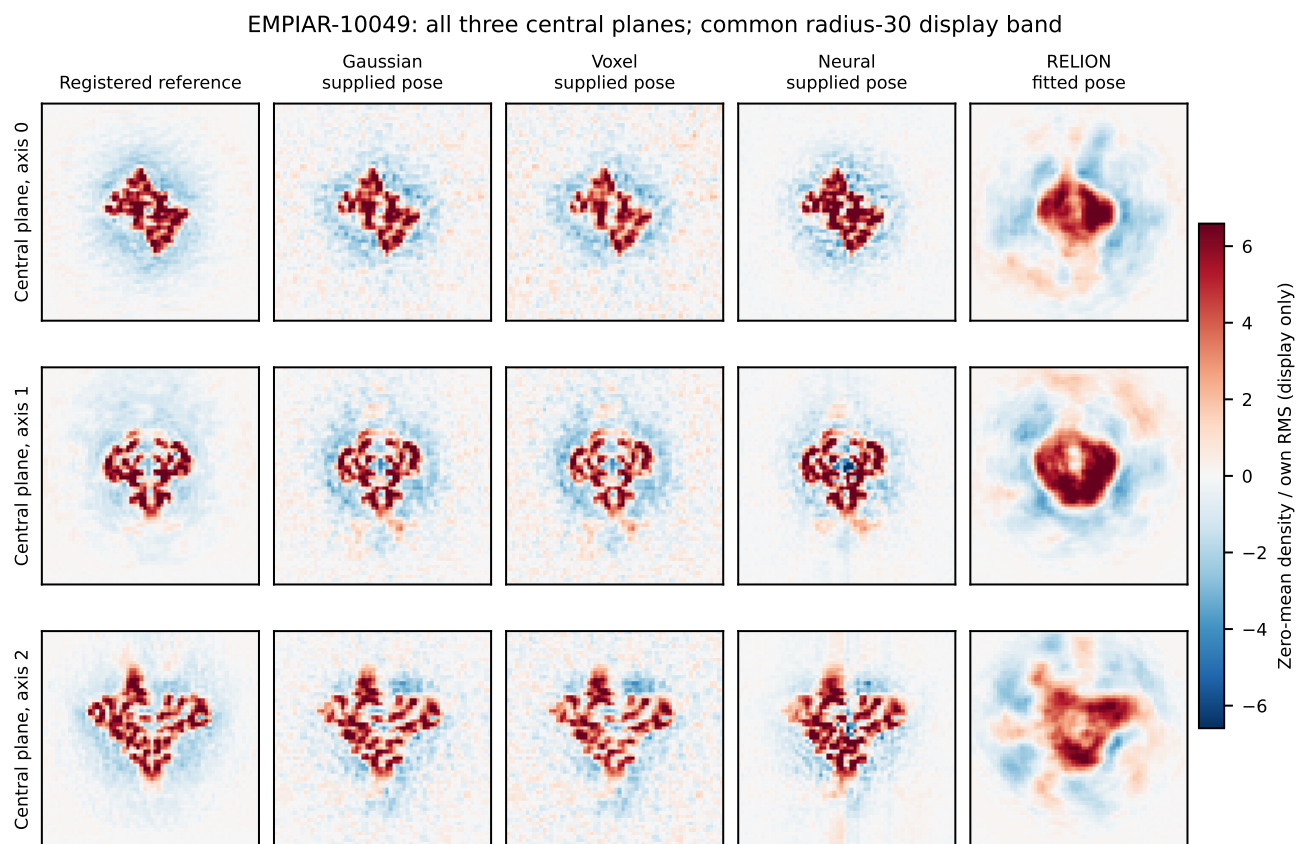


Figure 6. EMPIAR-10049: all three central planes of the registered reference and four existing reconstructions. Each square spans $236.16 \text{ \AA}$. The display removes DC, retains a common radius-30 Fourier band, and divides each map by its own volumetric RMS; a pooled 99.5th-percentile absolute value fixes the common symmetric color range. This display normalization hides amplitude differences and is not used for FSC or intervals. Views are fixed array planes, not selected for agreement. Gaussian, voxel and neural estimates average the locked halves; RELION uses its previously evaluated map. Registration is pilot-derived. The 10076 reference represents Class A, not the heterogeneous consensus.

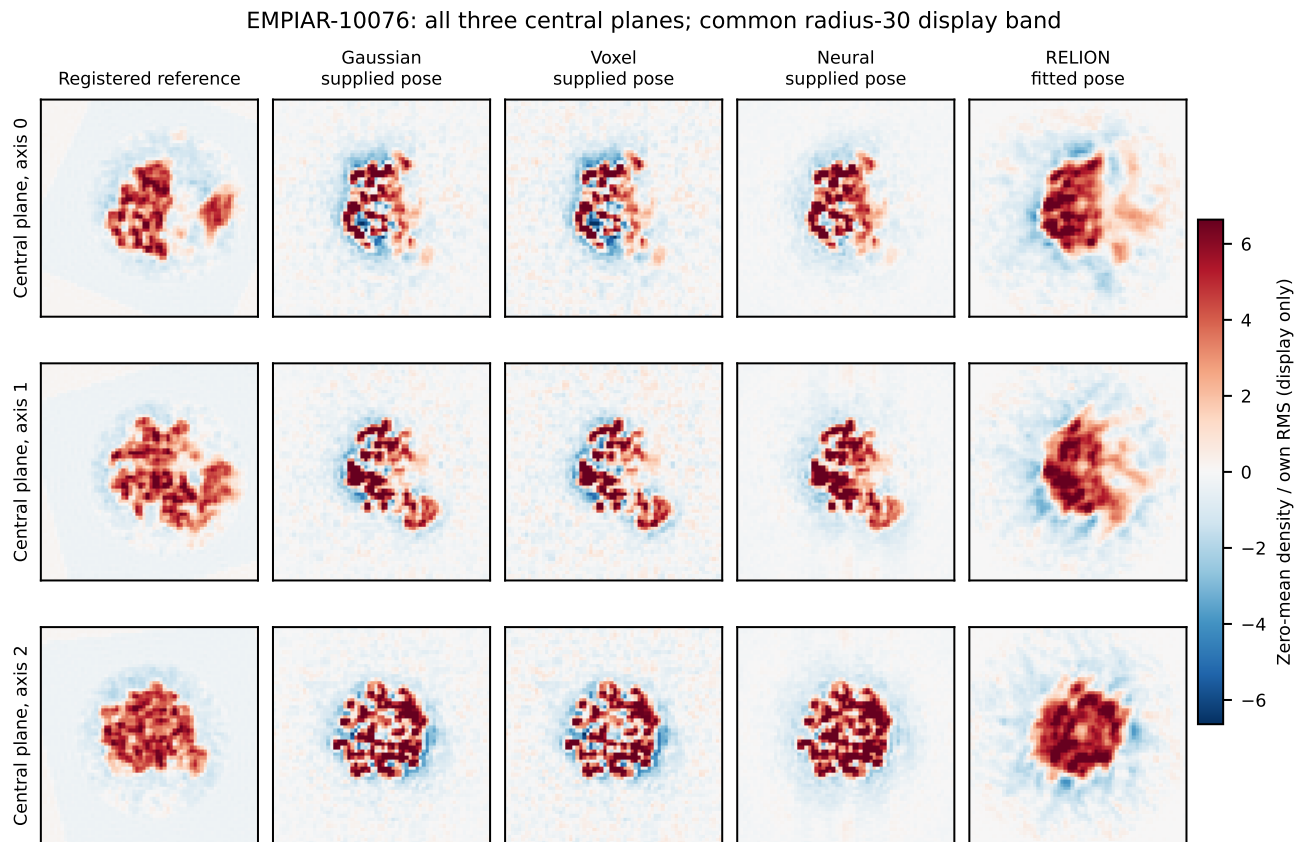


Figure 7. EMPIAR-10076: all three central planes of the registered reference and four existing reconstructions. Each square spans $419.2 \text{ \AA}$. The display removes DC, retains a common radius-30 Fourier band, and divides each map by its own volumetric RMS; a pooled 99.5th-percentile absolute value fixes the common symmetric color range. This display normalization hides amplitude differences and is not used for FSC or intervals. Views are fixed array planes, not selected for agreement. Gaussian, voxel and neural estimates average the locked halves; RELION uses its previously evaluated map. Registration is pilot-derived. The 10076 reference represents Class A, not the heterogeneous consensus.

F. Uncertainty Targets in the Recent Literature

The retrieval ledger contains 16,969 deduplicated broad-search records, mostly applications and not individually screened methods. The following distinctions guide baseline selection. They do not rank methods that solve different problems on one inappropriate score. The accompanying dated survey has 131 curated candidates across reconstruction, poses, heterogeneity, map validation, atomic ensembles and general inverse inference; its source manifest distinguishes retrieved material from targeted critical reading and inaccessible full text. This is a scoped critical survey rather than a claim to have exhaustively reviewed all cryo-EM applications. Versioned preprints are identified separately from published articles.

Additional local and atomic validation. The recent Gold-standard local toolbox compares half-map phase and amplitude, with explicit independent-noise assumptions in its SNR derivation (Vargas et al., 2026). EMMIVox combines molecular force fields, map likelihoods and half-map-informed uncertainty for atomic refinement (Hoff et al., 2024). Neither output is the fixed raw-particle density functional studied here.

Distributional inverse stability. The recent revision of Sanchez Espinosa et al. (2026) separates image-distribution discrepancy from latent Wasserstein error under assumed local stability. Its observability inequality bounds stability above, not below; it does not certify experimental coverage. A correctly specified Gaussian-mixture baseline wins its matched synthetic comparison, illustrating the importance of retaining favorable baselines.

Direct statistical uncertainty prior art. The probabilistic Fourier-slice model of Ullrich et al. (2020) is a close reconstruction-uncertainty comparator and explicitly recognizes model bias. Joint likelihood soft modes (Rangan et al., 2024) expose pose-density ambiguities. Lai et al. (2023) connect empirical Bayes and sequential MCMC filtering to dynamic image reconstruction, including scalar confidence sets and hybrid resampling. Their approximate fitted-family resampling and posterior computations differ from the present uniform conditional bounded-class statement; they cannot be dismissed as deterministic reconstruction methods. Neither general bias-aware inference nor the distinction between posterior variance and model error originates here.

Latent orientations and moment validation. Nonuniform moment recovery has explicit bandlimit and genericity conditions; numerical Jacobian ranks in unknown-law cases differ from exact uniqueness proofs (Sharon et al., 2020).

SubspaceMoM compresses moments to jointly recover structure and viewing laws, with its principal reconstruction tests on synthetic CTF-corrupted images (Hoskins et al., 2026). High-noise Fisher stratification for marginalized orbit likelihoods assumes Haar rotations, generic signals and projected identifiability conditions (Fan et al., 2024); it is not per-particle pose calibration. Moment-based structural comparison also profiles a viewing-law expansion (Zhang et al., 2024). Its 10076 example uses separate states and deposited shifts, with remaining acquisition effects and limited close-state discrimination. These are substantive alternatives to conditional local-pose analysis, without establishing the same finite-sample feature guarantee.

Adjacent radiative CT work also separates analytic Gaussian uncertainty from shared reconstruction error and distinguishes foreground evaluation from whole-volume scores (Zhao et al., 2026). Its density-moment formulas and validation distinctions further constrain novelty claims; its acquisition and estimands differ from cryo-EM.

Bayes’ Rays uses post-hoc displacement uncertainty for a frozen NeRF, with a diagonal curvature approximation (Goli et al., 2024). Its spatial target differs from an additive density interval. The dynamic extension proposed in Patel (2025, Section 5.2) is explicitly future work, with cryo-EM as an intended application; we do not treat that proposal as an experimentally validated density guarantee.

Pose inference versus pose confidence. Posterior-averaged orientation estimation (Xu et al., 2026) and probabilistic orientation search in SIMPLE (Van et al., 2025) improve pose estimation through different statistical or optimization constructions. Neither an orientation distribution nor a stochastic search neighborhood should be relabeled a calibrated per-particle confidence ball without verifying its guarantee. Our prescribed local balls avoid that claim, but consequently do not solve practical pose calibration.

Cryo-forum uses quaternion-distribution dispersion for error ranking and filtering (Chung, 2023). ARCHER learns reference-conditioned pose scores and evaluates cross-specimen transfer and junk rejection (Nguyen & Pham, 2026). CESPED standardizes experimental poses obtained by refinement, and explicitly acknowledges uncertainty in those labels (Sanchez-Garcia et al., 2024). CryoPARES transfers alignments between related experiments (Sanchez-Garcia et al., 2026); CryoFastAR combines multiview pose prediction, synthetic training and experimental fine-tuning (Zhang et al., 2025b). These approaches address efficiency and different transfer settings. Their quality scores and angular errors cannot supply a uniform maximum-error radius for every particle in a new stack. The 0.1° and 0.5° budgets in our frozen study are sensitivity settings, not experimentally established alignment accuracy.

Physical heterogeneity versus estimation error. RECOVAR estimates covariance and uses kernel regression to account for noisy latent coordinates (Gilles & Singer, 2025). DynaMight and 3DFlex represent motion (Schwab et al., 2024; Punjani & Fleet, 2023). Particle-counting work explicitly studies when assignments misestimate state populations (Evans et al., 2026). These motivate separate tests of population recovery, structural variation and uncertainty in a homogeneous density. The present bounded-density target covers only the last. CryoDRGN-AI addresses joint ab initio reconstruction (Levy et al., 2025); fixed-pose homogeneous training does not reproduce that task.

DynaMight’s held-out deformation disagreement and independent-half training assess motion errors; 3DFlex separates lower-frequency motion training from higher-frequency reconstruction validation. Their protocols must be examined at the level of shared components and evaluated frequencies. Mattingly et al. (2026) study how the image model limits learnable conformational populations, using information-based selection of ensemble granularity. That prior-averaged population target differs from our fixed homogeneous density functional. CryoBench likewise distinguishes representation learning, conformation assignment and volume recovery with several reference-based FSC metrics (Jeon et al., 2025). Our simulations posit one density per acquisition geometry; the actual stacks need not be homogeneous. In particular, the assembly-intermediate mixture in EMPIAR-10076 does not validate our shared-density assumption.

Map validation and learned priors. False-discovery confidence maps test density significance against a noise reference (Beckers et al., 2019). Directional resolution measures acquisition anisotropy (Vilas et al., 2020); independent-particle tests assess map fit (Ortiz et al., 2020). Bias/variance analyses explain why validation requires more than internal agreement (Sorzano et al., 2022). Blush introduces learned reconstruction regularization (Kimanius et al., 2024), while LocScale-2.0 attaches confidence to enhancement relative to its stated proxy (Bharadwaj et al., 2026). Compressed-image recovery with cryoSENSE targets acquisition fidelity rather than density-confidence coverage (Shabeeb et al., 2026). Matched evaluations must retain these targets. A sharper-looking map or a significant positive voxel is not itself a bound on error relative to unknown density.

Fixed-candidate ensemble analysis also separates discretization bias, statistical variability and early stopping (Mordant et al., 2026). Its simplified known-likelihood population problem differs from spatial density reconstruction with uncertain poses; it does not calibrate the nuisance bounds used here.

CryoRes predicts local resolution using deposited and

algorithm-derived training labels (Dai et al., 2023). Such supervision inherits the meaning of those labels; it does not transform a resolution score into a confidence interval. Similarly, held-out image prediction measures agreement with new observations, and can remain favorable when two reconstructions share unsupported density directions. Ground-truth simulation, new-exposure prediction, directional diagnostics and independent structural evidence therefore complement one another.

Invariant compression and thermodynamic targets. Moment-based posterior sampling conditions a learned prior on shift-invariant power spectra in one-dimensional orbit recovery (Janson & Andén, 2025). Compression discards phase information, so posterior concentration can rely on the prior. Separately, cryoTWIN/PaStEL targets conformational free energies under sampling, pose and equilibrium assumptions (Yamazaki et al., 2026). Neither problem is equivalent to our shared-density feature interval.

Atomic ensembles and independent modalities. Experiment-guided AlphaFold3 conditions structural generation on reconstructed ESP maps and other measurements (Maddipati et al., 2026). Its atomic-ensemble target differs from raw-particle density inference. The multimodal examples also illustrate why improved map agreement need not imply improved fit to a separate measurement modality. Independent evidence is relevant to assessing structural interpretations that the density measurements alone leave ambiguous.

Coordinate-error conformal prediction. CalPro calibrates protein-coordinate error scores at an exchangeable-protein level (Shihab et al., 2026). Its atomic target and calibration unit differ from a fixed density feature inferred from noisy projections. We do not interpret a calibrated protein-average score as simultaneous residue coverage or as experimental-density coverage.

Simultaneous inverse inference. Constrained test-inversion methods already address joint confidence regions for multiple functionals (Batlle et al., 2025). A single data-consistency region can support simultaneous projections, whereas separately calibrated marginal intervals generally require an appropriate correction. Our present scalar-target theorem is not a new general framework for simultaneous inference, and a colored map of marginal intervals must not be presented as simultaneous coverage.

Conformal calibration and its sampling unit. Posterior-variance calibration can provide error bounds even when a posterior sampler is approximate (Namhofer et al., 2024); its theorem uses independent identically distributed signal-observation pairs and conditions on a variance bin. QUTCC

learns quantile-dependent image predictions and calibrates marginal pixel errors from paired reference images (Ye et al., 2026). These methods can be useful without our fixed-signal radius assumption, but require a calibration population with the stated sampling relationship to deployment. Voxels or noise draws from one molecule are not automatically independent calibration molecules. Neither marginal population coverage nor variance-bin coverage is the same as coverage uniformly over every density in a specified class.

Prior-assisted alignment and atomic refinement. CryoPROS uses generated auxiliary particles for pose correction, with template-variation and noise-only controls (Zhang et al., 2025a). Such tests address model bias without calibrating experimental pose confidence regions. CoCoFold fine-tunes AlphaFold against particle images using a Gaussian-mixture forward model and supplied poses/CTFs (Liao et al., 2026). Its pose-perturbation checks support a sensitivity comparison, not a density-coverage guarantee. Both motivate independent validation of prior-assisted inference; neither is a matched density-functional uncertainty baseline here.

Open questions suggested by the survey. Particularly consequential gaps are absolute pose-confidence calibration under reference error; uncertainty in conformational populations when candidate structures are imperfect; validation of detail introduced by learned priors using independent particles; and uncertainty propagation through CTF fitting, image preprocessing, alignment and atomic interpretation. Acquisition chosen for a specific structural feature is another possible application of adjoint sensitivity. These are directions inferred from the literature, not priority claims or completed capabilities of this implementation. The present continuous audit addresses representation exclusion under stated nuisance assumptions; solving those assumptions empirically remains distinct work.

Alignment attenuation and conditioning. Classical alignment-error envelopes include orientation, location and other imaging errors (Jensen, 2001). Cross-validation against overfitting and noise correlation in three-dimensional alignment is also established (Shaikh et al., 2003). For these two historical papers our reading is limited to the primary author abstracts; we do not claim to have verified their full derivations. SIMPLE’s shared orientation-table updates (Van et al., 2025) illustrate that particle-wise local fits against a frozen template are not the same stochastic procedure as a full reconstruction pipeline. The phase control in Appendix D is therefore a diagnostic, not a discovery of alignment-induced bias.

Additional Gaussian and anisotropy context. Rendering-aware NIW posteriors explicitly depend

on surrogate statistics (Jia et al., 2026); photometric residual channels target view-dependent error prediction (Galappaththige et al., 2026). View-structured conformal prediction instead calibrates a within-view covered fraction using exchangeable labeled views (Chu et al., 2026). These distinguish posterior, ranking and coverage targets; none supplies raw-particle density calibration. A recent atomic-fitting preprint argues for anisotropic Fourier weighting (Yadgar & Lederman, 2026). Only its primary abstract was accessible in our reading, so its equivalence derivation has not been checked.

Processing-order validation. Post-acquisition super resolution uses movie preprocessing and perturbation controls alongside map/model agreement (Burton Smith & Murata, 2026). Those tests assess reconstruction fidelity; they do not establish the conditional noise law needed for density coverage.